

**Vidya Vikas Institute of Engineering and Technology,
Mysore
Library and Information Centre**



VTU Question Papers

ELCETRONICS & COMMUNICATION ENGINEERING

Odd Semester

2022-23

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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CBCS SCHEME

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18ES51

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023 Technological Innovation Management and Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the various functions of Management. (07 Marks)
- b. Describe the various roles of a Manager. (07 Marks)
- c. Distinguish between Management and Administration. Draw the diagram for time spent in administrative and managerial functions at different levels. (06 Marks)

OR

- 2 a. Define planning. Describe the importance of planning. (08 Marks)
- b. Explain all the steps in Rational Decision making with a neat diagram. (08 Marks)
- c. Give the differences between programmed and non-programmed decisions. (04 Marks)

Module-2

- 3 a. Explain the span of management concept its meaning and importance. (05 Marks)
- b. Define committee. Explain the different types of committees. (05 Marks)
- c. What is Recruitment? Explain the steps in the selection process. (10 Marks)

OR

- 4 a. Write about the requirements of effective direction. (08 Marks)
- b. Discuss Autocratic, Democratic and Free rein Leadership styles. (08 Marks)
- c. Explain the meaning of control. Write the benefits of control. (04 Marks)

Module-3

- 5 a. Explain the meaning of social responsibility. Describe the social responsibilities of Business towards consumer and community. (06 Marks)
- b. What is Social Audit? What are its benefits and limitations? (07 Marks)
- c. What is Corporate Governance? List the benefits of good Corporate Governance. (07 Marks)

OR

- 6 a. Define Entrepreneurship. Explain the different types of Entrepreneurs. (08 Marks)
- b. Explain Entrepreneurial development cycle. (07 Marks)
- c. List and explain any five characteristics of an entrepreneur. (05 Marks)

Module-4

- 7 a. Define Family Business. Explain the characteristics of a family owned business in India. (07 Marks)
- b. Explain the various types of family businesses. (07 Marks)
- c. Write the contributions of family businesses in India. Also explain the stages of development of family business. (06 Marks)

OR

- 8 a. Explain the ways of generating business ideas. (08 Marks)
b. Explain how to identify a business opportunity. (06 Marks)
c. Explain the concept of Financial Feasibilities. (06 Marks)

Module-5

- 9 a. What is the purpose of a Business plan? Explain. Also write the reasons for preparing a business plan. (07 Marks)
b. Why do some business plans fail? Explain. (07 Marks)
c. Explain the following terms:
i) Venture capital
ii) Angel Investing (06 Marks)

OR

- 10 a. Explain the six stages of venture capital financing as identified by Tyebjee and Bruno. (08 Marks)
b. Discuss the steps in PERT network analysis technique. What are its advantages? (08 Marks)
c. List differences between PERT and CPM. (04 Marks)

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18EC52

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Digital Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. State and prove circular time shift property. (06 Marks)
- b. Find the 4-point DFT of the sequence $x(n) = \cos\left(\frac{\pi n}{4}\right) + \sin\left(\frac{\pi n}{4}\right)$ use linearity property. (06 Marks)
- c. Consider 4-point sequences $x(n) = \cos\left(\frac{\pi n}{2}\right); 0 \leq n \leq 3$ (08 Marks)
- $h(n) = 2^n; 0 \leq n \leq 3$
- Compute circular convolution. Using concentric circle method. (06 Marks)

OR

- 2 a. State and prove Parseval's theorem. (06 Marks)
- b. Find 6-point DFT of the sequence $x(n) = n; 0 \leq n \leq 5$ (08 Marks)
- $= 0; \text{ otherwise}$
- c. Find the IDFT of the DFT $X(K) = \{6, -2 + j2, -2, -2 - j2\}$. (06 Marks)

Module-2

- 3 a. Consider a FIR with filter whose impulse response $h(n) = \{3, 2, 1, 1\}$ if the input is $x(n) = \{1, 2, 3, 3, 2, 1, 1, 2, 3, 5, 6, 1, 2, 0, 2, 1\}$, find the output using overlap add method assuming the length of block as 7. (10 Marks)
- b. Develop Radix-2 DIT-FFT algorithm and draw complete signal flow graph for $N = 8$. (10 Marks)

OR

- 4 a. Find the output $y(n)$ of a filter whose impulse response is $h(n) = \{1, 1, 1\}$ and the input signal to the filter is $x(n) = \{3, 1, 0, 1, 3, 2, 0, 1, 2, 1\}$. Using overlap save method. (10 Marks)
- b. First five points of the Eight point DFT of a real valued sequence is given by $x(0) = 0, x(3) = 2 - 2j$ (10 Marks)
- $x(1) = 2 + 2j, x(4) = 0$
- $x(2) = j4$
- Determine the remaining points. Hence find the original sequence $x(n)$ using Decimation in frequency FFT algorithm. (10 Marks)

Module-3

- 5 a. List the different types of windowing techniques used in the design of FIR filters. Write the analytical equations, draw the magnitude response and show the largest side lobe value below the dc magnitude. (08 Marks)
- b. The frequency response of an FIR filter is given by $H(\omega) = e^{-j3\omega} (1 + 1.8 \cos 3\omega + 1.2 \cos 2\omega + 0.5 \cos \omega)$ (06 Marks)
- Determine the coefficient of the impulse response $h(n)$ of the FIR filter.

- c. Determine the coefficient K_m of the lattice filter corresponding to FIR filter described by the system function $H(z) = 1 + 2z^{-1} + \frac{1}{3}z^{-2}$. And also draw the Lattice structure. (06 Marks)

OR

- 6 a. Determine the filter coefficient $h_d(n)$ for the desired frequency response of a Lowpass filter is given by
- $$H_d(w) = \begin{cases} e^{-j2w} & ; -\pi \leq w \leq \pi \\ 0 & ; \frac{\pi}{4} \leq w \leq \pi \end{cases}$$
- Find $h(n)$ and also frequency response $H(w)$ using Hamming window. (10 Marks)
- b. Obtain the cascade form realization of system function :
- $$H(z) = 1 + 5z^{-1} + 2z^{-2} + 2z^{-3}$$
- (05 Marks)
- c. Realize the following function in Direct form.
- $$H(z) = \left(1 + \frac{1}{2}z^{-1} + z^{-2} \right) \left(1 + \frac{1}{4}z^{-1} + z^{-2} \right)$$
- (05 Marks)

Module-4

- 7 a. Discuss the general procedure for IIR filter design using Bilinear transformation. (06 Marks)
- b. An analog filter is given by $H_a(s) = \frac{s+0.1}{(s+0.1)^2 + 16}$. Obtain digital IIR filter using bilinear transformation method. Digital filter is to have resonant frequency $\omega_r = \frac{\pi}{2}$ radians. (08 Marks)
- c. Compare FIR and IIR filter. (06 Marks)

OR

- 8 a. Design a Butterworth digital low pass filter with the following specifications.
- 3dB attenuation at the passband frequency of 1.5KHz
 - 10dB stopband attenuation at the frequency of 3KHz
 - Sampling frequency of 8000Hz.
- (10 Marks)
- b. A system is represented by a transfer function $H(z)$ is given by $H(z) = 1 + \frac{4z}{z - \frac{1}{2}} - \frac{2}{z - \frac{1}{4}}$
- Does this $H(z)$ represent a FIR or IIR filter? Why?
 - Draw direct form – I and Direct form – II realization by showing all differences equations?
- (10 Marks)

Module-5

- 9 a. Explain IEEE floating point formats using :
- Single precision format
 - Double precision format.
- (08 Marks)
- b. Discuss briefly multiplier and Accumulator unit in Digital signal processor hardware units. (04 Marks)
- c. Draw the block diagram to TMS320C3X floating point digital signal processor. (08 Marks)

OR

- 10 a. With block diagram explain Digital signal processor based on Harvard architecture. (06 Marks)
- b. Convert the Q-15 signed number to decimal numbers.
- 1.110101110000010
 - 0.100011110110010
- (04 Marks)
- c. Explain the basic architecture of TMS320CS54X used in fixed point Digital signal processor. (10 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Principles of Communication Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With proper necessary equations, explain the time domain and frequency domain expressions for an AM wave. Outline the waveforms and spectrum. (08 Marks)
- b. Explain the generation of AM wave using switching modulator with neat block diagram and relevant equations. (08 Marks)
- c. An audio frequency signal $10\sin 2\pi(500t)$ is used to amplitude modulate a carrier signal of $50\sin 2\pi(10^5 t)$. Assume modulation index as 0.2. Determine
 - (i) Side band frequencies.
 - (ii) Amplitude of each sideband.
 - (iii) Band width required. (04 Marks)

OR

- 2 a. What is a coherent detector used for DSB-SC? Why it is so named? Explain its working with a neat block diagram. What can be the problems in a coherent detector? (08 Marks)
- b. Compare and contrast standard AM, DSB-SC, SSB-SC and VSB-SC (at least 5 points in each) (05 Marks)
- c. When the modulation percentage is 75 an AM transmitter produces 10 kW. How much of this is carrier power? Determine the percentage of power saving if the carrier and one of the sidebands were suppressed before transmission took place. (07 Marks)

Module-2

- 3 a. From the fundamentals deduct an expression of WBFM and plot its frequency spectrum. (10 Marks)
- b. What is frequency modulation? Deduct the expression for a narrow band FM signal. Represent a narrow band FM signal with neat phasor diagram. (08 Marks)
- c. A FM signal has sinusoidal modulation with $W = 15$ kHz and modulation index $\beta = 2$. Using Carson's rule determine the transmission bandwidth and deviation ratio. Assume $\Delta f = 75$ kHz. (02 Marks)

OR

- 4 a. With relevant mathematical analysis and block diagrams show the reconstruction of message signal from FM wave using PLL. (10 Marks)
- b. Explain the generation of FM wave using a neat block diagram and necessary equations. (06 Marks)
- c. A Carrier is frequency modulated by a sinusoidal modulating signal of frequency 3 kHz resulting in a frequency deviation of 10 kHz.
 - (i) What is the bandwidth occupied by the modulated waveform?
 - (ii) If the amplitude of the modulating signal is increased by a factor of 2 and its frequency is lowered to 1 kHz. Determine the new bandwidth. (04 Marks)

Module-3

- 5 a. What is thermal Noise? List out different characteristics of thermal Noise. (06 Marks)
 b. What is white noise? Deduct the power spectral density and auto-correlation function for a RC-Low pass filtered white Noise. Also find the Noise equivalent bandwidth for the same and show its relationship with normal bandwidth. (08 Marks)
 c. Explain the applicability of pre-emphasis and de-emphasis with respect to FM system. (06 Marks)

OR

- 6 a. Determine the FOM for a DSB-SC receiver. (08 Marks)
 b. Determine the FOM for a standard AM receiver. (08 Marks)
 c. Explain capture effect and threshold effect with respect to FM receiver. (04 Marks)

Module-4

- 7 a. What are the advantages of digital modulation techniques over analog? (04 Marks)
 b. What is sampling theorem? Explain sampling with neat sketches and equations. What are the challenges faced with Nyquist criteria of sampling? (08 Marks)
 c. What is Flat top sampling with PAM. Explain the same with neat waveforms and derive the equation for flat-top sampled PAM. (08 Marks)

OR

- 8 a. What is multiplexing and why it is required in communication? Explain the working of TDM with neat block diagram. (08 Marks)
 b. What is pulse position modulation? Explain the generation of a PPM wave with neat block diagram and necessary waveforms. (08 Marks)
 c. What is aperture error in PAM? How to minimize it? (04 Marks)

Module-5

- 9 a. What is Quantization? Why it is required in digital communication? Explain symmetric quantizer of midtread and midrise type. (08 Marks)
 b. With neat block diagram, explain the working of PCM system. (08 Marks)
 c. What is companding? Explain different laws of companding. (04 Marks)

OR

- 10 a. What is Quantization noise? Derive the expression for O/P signal to Noise Ratio of a Quantizer. Consider a sinusoidal modulating signal of amplitude AM which uses all representation levels provided. Calculate the $(SNR)_O$ for the O/P of quantizer of the above signal. (08 Marks)
 b. What is Delta modulation? Explain the same with block diagrams. (06 Marks)
 c. Write a note on Vocadens. (06 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023 Information Theory & Coding

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the followings:
 - (i) Entropy
 - (ii) Information rate.
 - (iii) Self information.

(06 Marks)
- b. A binary source is emitting an independent sequence of 0's and 1's with probability of P and 1-P respectively. Plot the Entropy of this source versus P ($0 < P < 1$). (06 Marks)
- c. For the first order Markov statistical model shown in Fig. Q1 (c). Compute
 - (i) Probabilities of each state.
 - (ii) $H(s)$ and $H(s^{-2})$

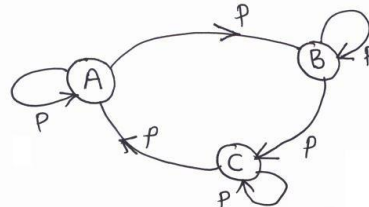


Fig. Q1 (c)

(08 Marks)

OR

- 2 a. For the first order Markoff model shown in Fig. Q2 (a). Find
 - (i) Entropy of each state.
 - (ii) Entropy of the source.
 - (iii) Prove that $G_1 \geq G_2 \geq H$

Assume $P(1) = P(2) = P(3) = \frac{1}{3}$

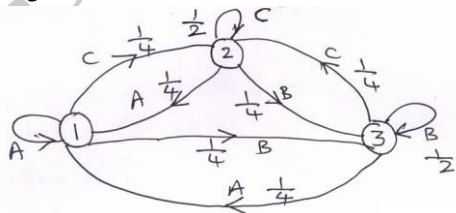


Fig. Q2 (a)

(12 Marks)

- b. The international Morse code uses a sequence of dots and dashes to transmit letters of the English alphabets. The dash represented by a current pulse that has a duration of 3 units and the dot has a duration of 1 unit. The probability of a dash is $\frac{1}{3}$ of the probability of occurrence of a dot.
 - (i) Calculate the information content of a dot and a dash.
 - (ii) Calculate $H(s)$ in the dot-dash code.
 - (iii) Assume that the dot lasts 1 msec.

Which is the same time interval as the pause between symbols? Find the average rate of information transmission.

(08 Marks)

Module-2

- 3 a. Construct a binary Shannon encoding algorithm for the following source with probabilities:
 $S = \{A, B, C, D, E\}$
 $P = \{0.4, 0.25, 0.15, 0.12, 0.08\}$
 Also compute the code Efficiency. (08 Marks)
- b. What is prefix of a code and explain with example. (04 Marks)
- c. Construct a Ternary code using Huffman Encoding algorithm for the source given with probabilities and move the composite symbol as low as possible.

Symbol :	A	B	C	D	E	F	G
Probabilities :	$\frac{1}{3}$	$\frac{1}{27}$	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{27}$	$\frac{1}{27}$

Also find the code efficiency.

(08 Marks)

OR

- 4 a. Check the following codes given in Table (1) are instantaneous or not with the help of KMI.

Symbols	Code A	Code B	Code C
A	0	0	00
B	10	11	01
C	110	100	10
D	1110	110	111
E	1111	1011	0110

Table (1)

(09 Marks)

- b. Design a source Encoder using Shannon encoding algorithm for the information source shown in Fig.Q4 (b). Compute the average output bit rate and efficiency of the code for $N = 1$. Assume $P_1 = P_2 = \frac{1}{2}$.

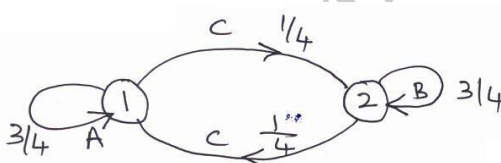


Fig. Q4 (b)

(11 Marks)

Module-3

- 5 a. Define the followings:
- Channel matrix.
 - Joint probability matrix.
 - Input entropy.
 - Output entropy.
- b. What is mutual information? Prove that $I(X, Y) \geq 0$.
- c. Determine the capacity of the channel shown in Fig. Q5 (c).

(08 Marks)

(08 Marks)

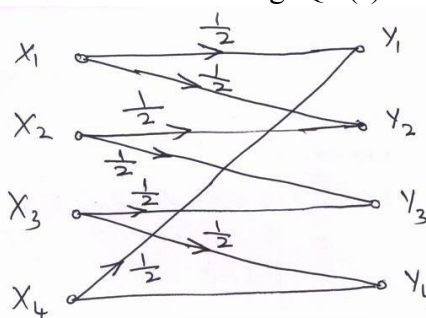


Fig. Q5 (c)

(04 Marks)

OR

- 6 a. Consider a channel matrix, $P\left(\frac{Y}{X}\right) = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$

with $P(X_1) = P(X_2) = P(X_3) = \frac{1}{3}$

Find $H(X)$, $H(Y)$, $H(X,Y)$, $H\left(\frac{Y}{X}\right)$ and $H\left(\frac{X}{Y}\right)$.

(08 Marks)

- b. The noise characteristic of a channel as shown in Fig. Q6 (b). Find the capacity of a channel using Muruga's method. Assume $\gamma_s = 1500$ symbols/sec.

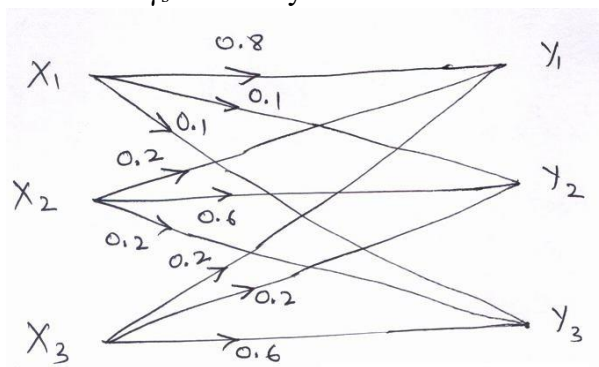


Fig. Q6 (b)

(08 Marks)

- c. Explain Binary Erasure channel.

(04 Marks)

Module-4

- 7 a. Define the following:

- (i) Hamming weight.
- (ii) Hamming distance.
- (iii) Minimum distance.

(06 Marks)

- b. For a (6, 3) linear block code, the parity matrix is,

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

- (i) Obtain the generator matrix.
- (ii) Write all possible code words.
- (iii) If the received code vector $R = 1\ 1\ 1\ 0\ 1\ 0$, detect and correct the single error.
- (iv) Draw the encoder and syndrome calculation block diagram.

(14 Marks)

OR

- 8 a. A Generator polynomial for a (15, 7) cyclic code is $g(x) = 1 + x^4 + x^6 + x^7 + x^8$.

- (i) Find the code vector for the message $D(x) = x^2 + x^3 + x^4$ using encoder circuit.
- (ii) Draw the syndrome calculation circuit and find the syndrome of the received polynomial.

$$z(x) = 1 + x + x^3 + x^6 + x^8 + x^9 + x^{11} + x^{14}$$

(16 Marks)

- b. Mention the advantages and disadvantages of error control coding.

(04 Marks)

Module-5

- 9 a. Consider the (3, 1, 2) convolution encoder with $g(1) = 110$, $g(2) = 101$ and $g(3) = 111$
- Draw the encoder diagram.
 - Find the code word for the message sequence (11101) using generator matrix/matrix method.
 - Find the code word for the message sequence (11101) using transform domain approach. (16 Marks)
- b. What are convolution codes? How it is different from block codes. (04 Marks)

OR

- 10 The (2, 1, 2) convolution encoder shown in Fig. Q10.
- Draw state transition table.
 - State diagram.
 - Draw the code tree and find the encoder output produced by the message (110)
 - Construct a Trellis diagram and find the encoder output produced by the message (110)

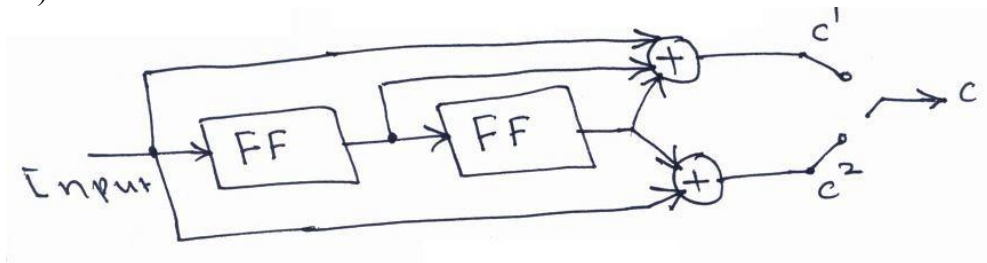


Fig. Q10

(20 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Electromagnetic Waves

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. The three vertices of a triangle are located at $A(6, -1, 2)$, $B(-2, 3, -4)$ and $C(-3, 1, 5)$. Find (i) $R_{AB} \times R_{AC}$ (ii) Area of triangle (04 Marks)
- b. Define Electric field intensity. Derive the expression for electric field intensity due to infinite line charge. (10 Marks)
- c. Given the electric flux density $\vec{D} = 0.3r^2 \hat{a}_r \text{ nC/m}^2$ in free space.
 - (i) Find E at point $P(r = 2, \theta = 25^\circ, \phi = 90^\circ)$.
 - (ii) Find total charge within the sphere $r = 3$.
 - (iii) Find total electric flux leaving the sphere $r = 4$. (06 Marks)

OR

- 2 a. Four identical 3 nC (nano Coulomb) charges are located at $P_1(1, 1, 0)$, $P_2(-1, 1, 0)$, $P_3(-1, -1, 0)$ and $P_4(1, -1, 0)$. Find the electric field intensity $\vec{E}$ at $P(1, 1, 1)$. (10 Marks)
- b. Infinite uniform line charges of 5 nC/m lie along the (positive and negative) x and y axes in free space. Find $\vec{E}$ at $P_A(0, 0, 4)$. (04 Marks)
- c. Define Coulomb's law. Make use of this to find the force on Q_1 . Given that the point charges $Q_1 = 50 \mu\text{C}$ and $Q_2 = 10 \mu\text{C}$ are located at $(-1, 1, -3) \text{ m}$ and $(3, 1, 0) \text{ m}$ respectively. (06 Marks)

Module-2

- 3 a. Explain Gauss law applicable to the case of infinite line charge and derive the relation used. (08 Marks)
- b. Evaluate both sides of the divergence theorem for the field $\vec{D} = 2xy \hat{a}_x + x^2 \hat{a}_y \text{ C/m}^2$ and the rectangular parallelepiped formed by the planes $x = 0$ and 1 , $y = 0$ and 2 and $z = 0$ and 3 . (08 Marks)
- c. Given the potential field $V = 2x^2y - 5z$ and point $P(-4, 3, 6)$. (i) Find potential V at P . (ii) Field intensity $\vec{E}$, (iii) Volume charge density ρ_v . (04 Marks)

OR

- 4 a. Compute the numerical value for $\text{div} \vec{D}$ at the point specified below:
 $\vec{D} = (2xyz - y^2) \hat{a}_x + (x^2z - 2xy) \hat{a}_y + x^2y \hat{a}_z \text{ C/m}^2$ at $P_A(2, 3, -1)$ (04 Marks)
- b. Show that Electric field is a negative gradient of potential. (08 Marks)
- c. Let $E = y \hat{a}_x \text{ V/m}$ at a certain instant of time and calculate the work required to move a 3 C charge from $(1, 3, 5)$ to $(2, 0, 3)$ along the straight line segment joining
 - (i) $(1, 3, 5)$ to $(2, 3, 5)$ to $(2, 0, 5)$ to $(2, 0, 3)$
 - (ii) $(1, 3, 5)$ to $(1, 3, 3)$ to $(1, 0, 3)$ to $(2, 0, 3)$ (08 Marks)

Module-3

- 5 a. Solve the Laplace's equation for the potential field in the homogenous region between the two concentric conducting spheres with radii 'a' and 'b' such that $b > a$, if potential $V = 0$ at $r = b$ and $V = V_0$ at $r = a$. Also find the capacitance between two concentric spheres. (10 Marks)
- b. State and explain Biot-Savart law applicable to magnetic field. (06 Marks)
- c. Calculate the value of vector current density in a rectangular coordinates at $P_A(2, 3, 4)$ if $\vec{H} = x^2 z \vec{a}_y - y^2 x \vec{a}_z$. (04 Marks)

OR

- 6 a. State and illustrate uniqueness theorem. (08 Marks)
- b. Define Stoke's theorem. Use this theorem to evaluate both sides of the theorem for the field $\vec{H} = 6xy \vec{a}_x - 3y^2 \vec{a}_y$ A/M and the rectangular path around the region, $2 \leq x \leq 5$, $-1 \leq y \leq 1$ $z = 0$. Let the positive direction of ds be $\vec{a}_z$. (12 Marks)

Module-4

- 7 a. Obtain the expression for magnetic force between differential current elements. (06 Marks)
- b. Derive the boundary conditions to apply to $\vec{B}$ and $\vec{H}$ at the interface between two different magnetic materials. (08 Marks)
- c. The point charge $q = 18 \text{ nC}$ has a velocity of $5 \times 10^6 \text{ m/s}$ in the direction $\vec{a}_v = 0.60 \vec{a}_x + 0.75 \vec{a}_y + 0.30 \vec{a}_z$. Calculate the magnitude of the force exerted on the charge by the field,
- (i) $\vec{B} = -3 \vec{a}_x + 4 \vec{a}_y + 6 \vec{a}_z$ mT
- (ii) $\vec{E} = -3 \vec{a}_x + 4 \vec{a}_y + 6 \vec{a}_z$ kV/m
- (iii) $\vec{B}$ and $\vec{E}$ acting together (06 Marks)

OR

- 8 a. Find the magnetization in a magnetic material, where
- (i) $\mu = 1.8 \times 10^{-5} \text{ H/m}$ and $H = 120 \text{ A/m}$
- (ii) $\mu_r = 22$, there are $8.3 \times 10^{28} \text{ atoms/m}^3$, and each atom has a dipole moment of $4.5 \times 10^{-27} \text{ A.m}^2$
- (iii) $B = 300 \mu\text{T}$ and $\chi_m = 15$. (06 Marks)
- b. Let permittivity be $5 \mu\text{H/m}$ in region A where $x < 0$ and $20 \mu\text{H/m}$ in region B, where $x > 0$. If there is a surface current density $\vec{K} = 150 \vec{a}_y - 200 \vec{a}_z$ A/m at $x = 0$, and if $\vec{H}_A = 300 \vec{a}_x - 400 \vec{a}_y + 500 \vec{a}_z$ A/m. Compute
- (i) $|H_{tA}|$ (ii) $|H_{nA}|$ (iii) $|H_{tB}|$ (iv) $|H_{nB}|$ (08 Marks)
- c. State and explain Faraday's law of electromagnetic induction. (06 Marks)

Module-5

- 9 a. List and explain Maxwell's equations in point and integral form. (08 Marks)

- b. The time domain expression for the magnetic field of a uniform plane wave travelling in free space is given by,

$$H(z,t) = \bar{a}_y 2.5 \cos(1.257 \times 10^9 t - K_0 z) \text{ mA/m.}$$

Compute

- (i) The direction of wave propagation.
 - (ii) Operating frequency
 - (iii) Phase constant.
 - (iv) The time domain expression for electric field $E(z,t)$ starting from the Maxwell's equations.
 - (v) The phasor form of both the electric and magnetic field. (10 Marks)
- c. For silver the conductivity is $\sigma = 3 \times 10^6 \text{ S/m}$. At what frequency will the depth of penetration be 1 mm. (02 Marks)

OR

- 10 a. State and explain Poynting theorem and write the equation both in point and integral form. (08 Marks)
- b. Simplify the value of K to satisfy the Maxwell's equations for region $\sigma = 0$ and $\rho_v = 0$ if $\bar{D} = 10x\bar{a}_x - 4y\bar{a}_y + kz\bar{a}_z \text{ } \mu\text{C/m}^2$ and $B = 2\bar{a}_y \text{ mT}$. (06 Marks)
- c. A plane wave of 16 GHz frequency and $E = 10 \text{ V/m}$ propagates through the body of salt water having constant $\epsilon_r = 100$, $\mu_r = 1$ and $\sigma = 100 \text{ s/m}$. Determine attenuation constant, phase constant, phase velocity and intrinsic impedance and depth and penetration. (06 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Verilog HDL

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain typical design flow for designing VLSI circuit using the flow chart. (08 Marks)
- b. i) A 4-bit ripple carry adder (Ripple – Add) contains four 1-bit full adders (FA). Define the module FA. Do not define the internals or the terminal list. Define the module Ripple – Add. Do not define the internals or the terminal list. Instantiate four full adder of the type FA in the module Ripple-Add and call them fa0, fa1, fa2, and fa3.
- ii) Define the module IS, using the module/endmodule keywords. Instantiate the modules MEM, Se, Xbar and call the instances mem1, se1 and Xbar 1, respectively. You do not need to define the internals. Assume that the module IS has no terminals. (06 Marks)
- c. What are the two styles of stimulus applications? Explain each method in brief. (06 Marks)

OR

- 2 a. Explain the trends in HDL. (04 Marks)
- b. With a hierarchical diagram of a 4-bit ripple carry counter, explain the design hierarchy (10 Marks)
- c. What is the difference between a module and a module instance? Explain with an example. (06 Marks)

Module-2

- 3 a. Describe different methods of connecting parts to internal signals. (06 Marks)
- b. Explain \$ display, \$ monitor, \$ finish and \$ stop system tasks with examples. (08 Marks)
- c. What are the basic components of a module? Explain all the components of a verilog module with a neat diagram. (06 Marks)

OR

- 4 a. Declare the following variables in verilog.
 - i) An 8-bit vector net called a – in
 - ii) A 16-bit hexadecimal unknown number with all x's
 - iii) A memory MEM containing 256 words of 64 bits each
 - iv) A parameter cache-size equal to 512. (04 Marks)
- b. With example explain different types of lexical conventions. (08 Marks)
- c. Write verilog description of SR latch. Also write stimulus code. (08 Marks)

Module-3

- 5 a. Write a verilog dataflow description for 4-bit full adder with carry lookahead. (06 Marks)
- b. What would be the output of the following
 $a = 4'b1010, b = 4'b1111$
 - i) $a \& b$ (ii) $a \& \& b$ (iii) $\& a$ (iv) $a \gg 1$ (v) $a \gg \gg 1$
 - (vi) $y = \{2\{a\}\}$ (vii) $a \wedge b$ (viii) $z = \{a, b\}$ (08 Marks)
- c. What re rise, fall and Turn-off delays? How they are specified in verilog? (06 Marks)

OR

- 6 a. A full subtractor has three 1-bit inputs x, y and z (previous borrow) and two 1-bit outputs D (Difference) and B (Borrow) the logic equations are

$$D = \overline{X}YZ + \overline{X}Y\overline{Z} + X\overline{Y}\overline{Z} + XYZ$$

$$B = \overline{X}Y + \overline{X}Z + YZ$$
 Write verilog description using dataflow modeling. Instantiate the subtractor inside a stimulus block and test all possible combinations of inputs X, Y and Z. (06 Marks)
- b. Discuss the And/or and Not gates with respect to logic symbols, gate instantiation and truth table. (06 Marks)
- c. Design AND-OR-INVERT (AOI) based 4:1 multiplexer write verilog description for the same and its stimulus. (08 Marks)

Module-4

- 7 a. Explain the following assignment statements and non-blocking assignment statements with relevant examples. (06 Marks)
- b. Write a verilog program for 8-to-1 multiplexer using case statement. (08 Marks)
- c. Give the differences between tasks and functions. (06 Marks)

OR

- 8 a. Explain sequential and parallel blocks with examples. (06 Marks)
- b. Design a negative edge-triggered D-flipflop (DUFF) with synchronous clear, active high (D-FF clears only at a negative edge of clock when clear is high). Design a clock with a period of 10 units and test the D-flipflop. (08 Marks)
- c. Write verilog program to call a function called calc-parity which computes the parity of a 32-bit data, [31-0] Data and display odd or even parity message. (06 Marks)

Module-5

- 9 a. Write a note on :
 i) Force and release
 ii) Defparam statement
 iii) time scale
 iv) file output. (08 Marks)
- b. Write a note on verification of gate level netlist. (04 Marks)
- c. With a neat flow chart explain computer Aided logic synthesis process. (08 Marks)

OR

- 10 a. What is logic synthesis? (04 Marks)
- b. Interpret the following verilog constructs after logic synthesis.
 i) The assign statement
 ii) The if-else statement
 iii) The case statement
 iv) The always statement (10 Marks)
- c. Write RTL description for magnitude comparator. (06 Marks)

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Question Paper Version : A

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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-
1. Which of the following is a possible producer in an ecosystem?
a) Animal b) Plants c) Human beings d) Fish
 2. The largest reservoir of nitrogen in our planet is
a) Oceans b) Biosphere c) Atmosphere d) Rivers
 3. India has the world's largest share of which of the following
a) Manganese b) Mica c) Copper d) Diamond
 4. Identify the non renewable source of energy from the following:
a) Coal b) Fuel cells c) Wind power d) Wave power
 5. Which of the following terminologies is not associated with the vertical structure of forest?
a) Canopy b) Understory c) Forest floor d) First floor
 6. Which of the following is cause of loss of biodiversity?
a) Habitat degradation b) Invasion of non-native species
c) Pollution d) All of these
 7. Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) Cyclone separator
c) Wet collector d) Catalytic converter
 8. When the solid waste consists of large amount of organic matter and if the moisture content is high, which of the following methods of treatment will be ideal?
a) Composting b) Palletizing
c) Incineration d) Recycling

9. Chernobyl Nuclear Disaster occurred in the year
a) 1984 b) 1985 c) 1986 d) 1987
10. The primary cause of acid rain around the world is
a) Carbon dioxide b) Sulphur dioxide
c) Carbon monoxide d) Ozone
11. World Environmental day is held every year on
a) June 5th b) October 2nd c) April 22nd d) November 1st
12. Ozone layer thickness is measured in _____.
a) mm b) cm c) Dobson unit d) Db
13. First of the major environmental protection acts to be promulgated in India was
a) The Water Act b) The Air Act
c) The Environment Act d) Noise Pollution Rules
14. Blue baby syndrome is caused due to _____.
a) Manganese b) Ozone c) Silver d) Nitrate
15. World Earth's day is annually celebrated on
a) April 22nd b) June 5th c) January 1st d) May 1st
16. The most important fuel used by nuclear power plant is
a) U-235 b) U-238 c) U-245 d) U-248
17. Which of the following is a biotic component of ecosystem?
a) Fungi b) Solar light
c) Temperature d) Humidity
18. Abiotic component includes
a) Soil b) Temperature
c) Water d) All of these
19. The word "Environment" is derived from
a) Greek b) French
c) Spanish d) English
20. Which of the following is absorbed by green plants from the atmosphere?
a) Carbon dioxide b) Water
c) Nutrients d) All of these
21. Eutrophication is
a) an improved quality of water in lakes
b) a process in carbon cycle
c) the result to accumulation of plant nutrients in water bodies
d) a water purification technique
22. Primary consumer is
a) Herbivores b) Carnivores
c) Macro consumers d) Omnivores

23. Which among the following is a climatic factor?
a) pressure b) humidity
c) temperature d) all of these
24. Biodiversity can be broadly classified into how many types?
a) 2 b) 5 c) 3 d) 4
25. Hot spot areas have
a) Low density of biodiversity b) Only endangered plants
c) High density of hot springs d) High density of biodiversity
26. About _____% of the earth's surface is covered by water.
a) 53% b) 19% c) 71% d) 90%
27. Deforestation means
a) preservation of forests b) destruction of forests
c) monocrop cultivation d) agriculture
28. When did National Disaster Management Authority formed?
a) 2000 b) 2005 c) 2010 d) 2015
29. Disaster is an event arising out of
a) result of hazard event b) causes of hazard event
c) causes of disaster event d) all of these
30. The scientific study of earthquake is called
a) seismograph b) seismology
c) both a and b d) none of these
31. South Africa is leading exporter of which mineral?
a) Copper b) Diamond
c) Silver d) Gold
32. The word 'sustainable development' came into existence in the year.
a) 1992 b) 1978 c) 1980 d) 1987
33. The other word of landscaping is
a) Reduction b) Restoration
c) Removing topsoil d) Restore
34. Cloud seeding with silver iodide is based on the
a) Bergeron process b) Collision-coalescence process
c) Both a and b d) None of these
35. Environmental pollution is due to
a) Rapid urbanization b) Deforestation
c) Afforestation d) a and b
36. The liquid waste from bathroom and kitchen is called
a) Sullage b) Domestic sewage
c) Storm water d) Runoff

37. BOD means
 a) Biochemical Oxygen Demand
 b) Chemical oxygen demand
 c) Biophysical Oxygen Demand
 d) All of these
38. Which of the following source is surface water?
 a) Springs
 b) Streams
 c) Deep wells
 d) All of these
39. Which of the following is biodegradable?
 a) Plastics
 b) Domestic sewage
 c) Detergents
 d) a and c
40. Blaring sounds known to cause
 a) Mental distress
 b) High cholesterol
 c) Neurological problems
 d) All of these
41. "Minamata Disease" is caused due to
 a) Lead
 b) Arsenic
 c) Mercury
 d) Cadmium
42. Alternative eco-friendly fuel for automobiles is
 a) Petrol
 b) Diesel
 c) CNG
 d) Kerosene
43. Population explosion will cause
 a) Biodiversity
 b) Stress on ecosystem
 c) More employment
 d) None of these
44. Which of the following is having high population density?
 a) India
 b) China
 c) USA
 d) Western Europe
45. Demography is the study of
 a) Animals behaviour
 b) Population growth
 c) River
 d) None of these
46. Forest are called as _____.
 a) Air purifier
 b) Earth's lungs
 c) Oxygen reservoir
 d) CO₂ absorbers
47. Which of the following is the facility that the urban people enjoy?
 a) Better quality of air
 b) Better communication access
 c) Large land at cheap rates
 d) None of these
48. Which of the following is an air pollutant?
 a) Carbon dioxide
 b) Oxygen
 c) Nitrogen
 d) Particulate matter
49. Cyoto toxic and expired drugs are disposed of by
 a) damping
 b) autoclave
 c) incineration
 d) chemical disinfection
50. The colour code of plastic bag for disposing of microbial laboratory culture waste.
 a) Black
 b) Red
 c) Blue
 d) White

51. Which of the following element make e-waste hazardous in nature?

a) Land	b) Glass
c) Plastic	d) Iron
52. What is the hazardous pollutant released from batteries?

a) Arsenic	b) Barium
c) Cobalt	d) Cadmium
53. What is the term used for reuse of sewage sludge?

a) Compost	b) Solids
c) Biosolids	d) Sludge
54. Reduction in brightness of the famous Taj Mahal is due to

a) Global warming	b) Air pollution
c) Ozone depletion	d) Afforestation
55. E.I.A. can be expanded as

a) Environment and Industrial Act	b) Environment of Impact Activities
c) Environmental Impact Assessment	d) Environmentally Important Activity
56. Organic Farming is

a) Farming without using pesticides and chemical fertilizer
b) Enhances biodiversity
c) Promotes soil biological activity
d) All of these
57. Bio-remediation means the removal of contaminants from

a) Soil	b) Wastewater
c) Groundwater	d) Both soil and ground water
58. Plants use _____ gas for photosynthesis.

a) Oxygen	b) Methane
c) Nitrogen	d) Carbon dioxide
59. What is the maximum allowable concentration of fluorides in drinking water?

a) 1.0 mg/l	b) 1.25 mg/l
c) 1.50 mg/l	d) 1.75 mg/l
60. Forest rich area in Karnataka is found in _____.

a) Western Ghats	b) Bandipur
c) Nagarhole	d) Mangalore
61. Among the fresh water available in the earth the percentage of surface water is about

a) 50%	b) 10%
c) 5%	d) less than 1%
62. Hepatitis is caused by

a) Protozoa	b) Virus
c) Bacteria	d) Fungus

63. In India groundwater resources are rich in
 a) Plains of river Kaveri and Krishna b) The Deccan plateau
 c) The Gangetic plains d) The plains of Netravati and Kapila
64. The required iron content in drinking water as specified by BIS is
 a) 300 mg/l b) 30 mg/l c) 3 mg/l d) 0.30 mg/l
65. Molasses from sugar industry is used to generate
 a) Biodiesel b) Hydrogen
 c) Bioethanol d) Biomethanol
66. Wind Farms are located in
 a) River basin b) Plain area
 c) Hilly area d) Valley area
67. Biomass consists of
 a) Lignin b) Hemi cellulose
 c) Cellulose d) All of these
68. Natural gas contains
 a) Carbon dioxide b) Hydrogen
 c) Methane d) Nitrogen
69. Anti tobacco day is mentioned on
 a) 31st May b) 30st June
 c) 31st July d) 31st August
70. Population explosion will cause
 a) Socio-Economic Problems b) Food Scarcity
 c) Energy crises d) All of these
71. In geosynchronous orbit altitude of the satellite is about
 a) 36,000 kms b) 10,000 kms
 c) 50,000 kms d) None of these
72. The Air (Prevention and Control of Pollution) Act was enacted in the year.
 a) 1987 b) 1981 c) 1991 d) 1988
73. Kudremukh Iron ore mine, Karnataka was closed due to
 a) River pollution and threat to biodiversity
 b) Land encroachment
 c) Radioactive hazards
 d) Serious health hazard
74. On the eve of Gandhi Jayanthi which andolan was launched by our Honorable Prime Minister
 a) Swadeshi b) Sarvashikshana Abhiyana
 c) Suvarnagrama d) Swach Bharath
75. An international agreement signed in the year 1987, to protect stratospheric ozone is known as
 a) Montreal protocol b) Kyoto protocol
 c) Earth summit d) None of these

76. The explosion of First Atomic Bomb was done in Hiroshima and Nagasaki in
 a) 1946 b) 1986 c) 1945 d) 1947
77. A dangerous pesticide which has been reported to cause physical deformities to people of Kerala and Karnataka states
 a) Endosulfan b) Fluorides c) DDT d) Dioxygene
78. Visible portion of EMR ranges between
 a) 0.4 - 0.76 μm b) 10.5 - 12.5 μm c) 8.0 - 14.0 μm d) None of these
79. Data representation in Raster data is by
 a) pixel b) points, lines and polygon
 c) latitude and longitude d) none of these
80. In water treatment, alum is used for
 a) softening b) coagulation c) filtration d) disinfection
81. GIS stands for
 a) Geostationary Interact Sector b) Geographical Information System
 c) Geotechnical Information Society d) Geothermal Investigation Site
82. LPG is a mixture of
 a) N_2 and H_2S b) CO_2 and N_2
 c) Propane and butanes d) Methane and CO_2
83. The Tiger Conservation Project was started in
 a) 1973 b) 1975 c) 1981 d) 2000
84. The leader of "Chipko Movement" is
 a) Sunderlal Bahuguna b) Medha Patkar
 c) Vandana Shiva d) Mahatma Gandhi
85. Which of the following is the source of Fly-ash?
 a) Vehicular exhaust b) Sewage
 c) Thermal power plant d) All of these
86. The permissible range of pH for drinking water as per the Indian Standard is
 a) 6 to 9 b) 6.5 to 8.5
 c) 6 to 8.5 d) 6.5 to 7.5
87. Water logging is a phenomena in which
 a) Water patterns are rotated
 b) Soil root zone becomes saturated due to over irrigation
 c) Erosion of soil
 d) Soil degradation
88. Carbon content is higher in
 a) Living matter b) Soil
 c) Water d) Atmosphere
89. Springs means
 a) Surface water b) Atmosphere water
 c) Both (a) and (b) d) Ground water

- a) Radiation means deliberately introducing micro organisms to
 b) Waste water
 c) Both soil and ground water
 d) Both soil and ground water
- a) Karnataka State Pollution Control Board (KSPCB) was established in
 b) 1982
 c) 1973
 d) 1975
- a) Which of the following is not a part of the hydrological cycle?
 b) Infiltration
 c) Evaporation
 d) Perspiration
- a) The International Earth Summit was held at
 b) Russia
 c) Rio de Janeiro
 d) Johannesburg
- a) Among the following has highest percentage of calorific value
 b) Peat
 c) Lignite
 d) Bituminous coal
- a) Nitrogen fixing bacteria exists in
 b) Stem
 c) Root
 d) Flower
- a) The major components of ecosystem are
 b) Ecologic and climatic
 c) Abiotic and biotic
 d) Abiotic and biotic
- a) Solar energy is a
 b) Wind energy
 c) Geothermal energy
 d) Solar energy
- a) The average life expectancy around the world is currently
 b) Increasing
 c) Decreasing
 d) Not changing
- a) The Universal declaration of Human Rights was proclaimed by the
 b) 1947
 c) 1948
 d) 1949
- a) The objective of Integrated Child Development Service (ICDS) are
 b) Health check up and
 c) School non formal education
 d) All of these
- * * * * *

radiation means deliberately introducing micro organisms to
 a) Water b) Waste water
 c) Air d) Both soil and ground water

Karnataka State Pollution Control Board (KSPCB) was established in
 a) 1980 b) 1982
 c) 1973 d) 1975

Which of the following is not a part of the hydrological cycle?
 a) Condensation b) Infiltration
 c) Precipitation d) Perspiration

The 1992 International Earth Summit was held at
 a) Rio de Janeiro b) Russia
 c) London d) Johannesburg

Among the following has highest percentage of calorific value
 a) Lignite b) Peat
 c) Coal d) Bituminous coal

Nitrogen fixing bacteria exists in
 a) Root b) Stem
 c) Leaf d) Flower

The major components of ecosystem are
 a) Abiotic and isotropic b) Ecologic and climate
 c) Abiotic and biotic d) Abiotic and isotropic

Solar energy is a
 a) Renewable energy b) Wind energy
 c) Non renewable energy d) Solar energy

The average life expectancy around the world is currently
 a) Increasing b) Increasing
 c) Decreasing d) Not changing

The Universal Declaration of Human Rights was proclaimed by the
 a) 1945 b) 1947
 c) 1948 d) 1949

The Integrated Child Development Service (ICDS) is a
 a) Health check up b) Health check up and
 c) School non formal education d) All of these

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Question Paper Version : B

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
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-
1. The Karnataka State Pollution Control Board (KSPCB) was established in the year.
a) 1974 b) 1982 c) 1973 d) 1983
 2. Which of the following is not a part of the hydrological cycle?
a) Precipitation b) Infiltration
c) Transpiration d) Perspiration
 3. First International Earth Summit was hold at
a) USA b) Russia
c) Rio-de-Janerio d) Johannesburg
 4. Which among the following has highest percentage of calorific value?
a) Anthracite b) Peat
c) Lignite d) Bituminous coal
 5. Nitrogen fixing bacteria exists in
a) Leaf b) Stem
c) Roots d) Flower
 6. The two major components of ecosystem are
a) Adiabatic and isotropic b) Ecologic and climatologic
c) Cyclic and biologic d) Abiotic and biotic
 7. Geothermal energy is a
a) Heat energy b) Wind energy
c) Current energy d) Solar energy

21. In geosynchronous orbit altitude of the satellite is about
 - a) 36,000 kms
 - b) 10,000 kms
 - c) 50,000 kms
 - d) None of these
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a) Reduction b) Restoration
c) Removing topsoil d) Restore
64. Cloud seeding with silver iodide is based on the
a) Bergeron process b) Collision-coalescence process
c) Both a and b d) None of these
65. Environmental pollution is due to
a) Rapid urbanization b) Deforestation
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a) Biochemical Oxygen Demand b) Chemical oxygen demand
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c) Detergents d) a and c
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a) an improved quality of water in lakes
b) a process in carbon cycle
c) the result to accumulation of plant nutrients in water bodies
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72. Primary consumer is
a) Herbivores b) Carnivores
c) Macro consumers d) Omnivores
73. Which among the following is a climatic factor?
a) pressure b) humidity
c) temperature d) all of these
74. Biodiversity can be broadly classified into how many types?
a) 2 b) 5 c) 3 d) 4

75. Hot spot areas have
 a) Low density of biodiversity
 b) Only endangered plants
 c) High density of hot springs
 d) High density of biodiversity
76. About _____% of the earth's surface is covered by water.
 a) 53%
 b) 19%
 c) 71%
 d) 90%
77. Deforestation means
 a) preservation of forests
 b) destruction of forests
 c) monocrop cultivation
 d) agriculture
78. When did National Disaster Management Authority formed?
 a) 2000
 b) 2005
 c) 2010
 d) 2015
79. Disaster is an event arising out of
 a) result of hazard event
 b) causes of hazard event
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 d) all of these
80. The scientific study of earthquake is called
 a) seismograph
 b) seismology
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 d) none of these
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 a) mm
 b) cm
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 a) The Water Act
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84. Blue baby syndrome is caused due to _____
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 b) Ozone
 c) Silver
 d) Nitrate
85. World Earth's day is annually celebrated on
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86. The most important fuel used by nuclear power plant is
 a) U-235
 b) U-238
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87. Which of the following is a biotic component of ecosystem?
 a) Fungi
 b) Solar light
 c) Temperature
 d) Humidity
88. Abiotic component includes
 a) Soil
 b) Temperature
 c) Water
 d) All of these
89. The word "Environment" is derived from
 a) Greek
 b) French
 c) Spanish
 d) English

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Question Paper Version : C

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. "Minamata Disease" is caused due to
a) Lead b) Arsenic c) Mercury d) Cadmium
 2. Alternative eco-friendly fuel for automobiles is
a) Petrol b) Diesel c) CNG d) Kerosene
 3. Population explosion will cause
a) Biodiversity b) Stress on ecosystem
c) More employment d) None of these
 4. Which of the following is having high population density?
a) India b) China c) USA d) Western Europe
 5. Demography is the study of
a) Animals behaviour b) Population growth
c) River d) None of these
 6. Forest are called as _____.
a) Air purifier b) Earth's lungs c) Oxygen reservoir d) CO₂ absorbers
 7. Which of the following is the facility that the urban people enjoy?
a) Better quality of air b) Better communication access
c) Large land at cheap rates d) None of these
 8. Which of the following is an air pollutant?
a) Carbon dioxide b) Oxygen
c) Nitrogen d) Particulate matter

9. Cyoto toxic and expired drugs are disposed of by
 - a) damping
 - b) autoclave
 - c) incineration
 - d) chemical disinfection
10. The colour code of plastic bag for disposing of microbial laboratory culture waste.
 - a) Black
 - b) Red
 - c) Blue
 - d) White
11. Among the fresh water available in the earth the percentage of surface water is about
 - a) 50%
 - b) 10%
 - c) 5%
 - d) less than 1%
12. Hepatitis is caused by
 - a) Protozoa
 - b) Virus
 - c) Bacteria
 - d) Fungus
13. In India groundwater resources are rich in
 - a) Plains of river Kaveri and Krishna
 - b) The Deccan plateau
 - c) The Gangetic plains
 - d) The plains of Netravati and Kapila
14. The required iron content in drinking water as specified by BIS is
 - a) 300 mg/l
 - b) 30 mg/l
 - c) 3 mg/l
 - d) 0.30 mg/l
15. Molasses from sugar industry is used to generate
 - a) Biodiesel
 - b) Hydrogen
 - c) Bioethanol
 - d) Biomethanol
16. Wind Farms are located in
 - a) River basin
 - b) Plain area
 - c) Hilly area
 - d) Valley area
17. Biomass consists of
 - a) Lignin
 - b) Hemi cellulose
 - c) Cellulose
 - d) All of these
18. Natural gas contains
 - a) Carbon dioxide
 - b) Hydrogen
 - c) Methane
 - d) Nitrogen
19. Anti tobacco day is mentioned on
 - a) 31st May
 - b) 30st June
 - c) 31st July
 - d) 31st August
20. Population explosion will cause
 - a) Socio-Economic Problems
 - b) Food Scarcity
 - c) Energy crises
 - d) All of these
21. GIS stands for
 - a) Geostationary Interact Sector
 - b) Geographical Information System
 - c) Geotechnical Information Society
 - d) Geothermal Investigation Site
22. LPG is a mixture of
 - a) N₂ and H₂S
 - b) CO₂ and N₂
 - c) Propane and butanes
 - d) Methane and CO₂

23. The Tiger Conservation Project was started in
 a) 1973 b) 1975 c) 1981 d) 2000
24. The leader of “Chipko Movement” is
 a) Sunderlal Bahuguna b) Medha Patkar
 c) Vandana Shiva d) Mahatma Gandhi
25. Which of the following is the source of Fly-ash?
 a) Vehicular exhaust b) Sewage
 c) Thermal power plant d) All of these
26. The permissible range of pH for drinking water as per the Indian Standard is
 a) 6 to 9 b) 6.5 to 8.5
 c) 6 to 8.5 d) 6.5 to 7.5
27. Water logging is a phenomena in which
 a) Water patterns are rotated
 b) Soil root zone becomes saturated due to over irrigation
 c) Erosion of soil
 d) Soil degradation
28. Carbon content is higher in
 a) Living matter b) Soil
 c) Water d) Atmosphere
29. Springs means
 a) Surface water b) Atmosphere water
 c) Both (a) and (b) d) Ground water
30. Bio-remediation means deliberately introducing micro organisms to break pollutants.
 a) Soil b) Waste water
 c) Ground water d) Both soil and groundwater
31. Which of the following is a possible producer in an ecosystem?
 a) Animal b) Plants c) Human beings d) Fish
32. The largest reservoir of nitrogen in our planet is
 a) Oceans b) Biosphere c) Atmosphere d) Rivers
33. India has the world’s largest share of which of the following
 a) Manganese b) Mica c) Copper d) Diamond
34. Identify the non renewable source of energy from the following:
 a) Coal b) Fuel cells c) Wind power d) Wave power
35. Which of the following terminologies is not associated with the vertical structure of forest?
 a) Canopy b) Understory c) Forest floor d) First floor
36. Which of the following is cause of class of biodiversity?
 a) Habitat degradation b) Invasion of non-native species
 c) Pollution d) All of these

37. Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) Cyclone separator
c) Wet collector d) Catalytic converter
38. When the solid waste consists of large amount of organic matter and if the moisture content is high, which of the following methods of treatment will be ideal?
a) Composting b) Palletizing
c) Incineration d) Recycling
39. Chernobyl Nuclear Disaster occurred in the year
a) 1984 b) 1985 c) 1986 d) 1987
40. The primary cause of acid rain around the world is
a) Carbon dioxide b) Sulphur dioxide
c) Carbon monoxide d) Ozone
41. Eutrophication is
a) an improved quality of water in lakes
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a) June 5th b) October 2nd c) April 22nd d) November 1st
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a) mm b) cm c) Dobson unit d) Db
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a) U-235 b) U-238 c) U-245 d) U-248
57. Which of the following is a biotic component of ecosystem?
a) Fungi b) Solar light
c) Temperature d) Humidity
58. Abiotic component includes
a) Soil b) Temperature
c) Water d) All of these
59. The word "Environment" is derived from
a) Greek b) French
c) Spanish d) English
60. Which of the following is absorbed by green plants from the atmosphere?
a) Carbon dioxide b) Water
c) Nutrients d) All of these
61. The Karnataka State Pollution Control Board (KSPCB) was established in the year.
a) 1974 b) 1982 c) 1973 d) 1983
62. Which of the following is not a part of the hydrological cycle?
a) Precipitation b) Infiltration
c) Transpiration d) Perspiration
63. First International Earth Summit was held at
a) USA b) Russia
c) Rio-de-Janeiro d) Johannesburg
64. Which among the following has highest percentage of calorific value?
a) Anthracite b) Peat
c) Lignite d) Bituminous coal
65. Nitrogen fixing bacteria exists in
a) Leaf b) Stem
c) Roots d) Flower

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b) Ecologic and climatologic
c) Cyclic and biologic
d) Abiotic and biotic
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a) Heat energy
b) Wind energy
c) Current energy
d) Solar energy
68. The average life expectancy around the world is currently.
a) Decreasing
b) Increasing
c) Stabilizing
d) Not changing
69. The universal declaration of Human Rights was proclaimed by the UN in the year.
a) 1946
b) 1947
c) 1948
d) 1949
70. The objective of Integrated Child Development Service (ICDS) are
a) Immunization
b) Health check up and referral services
c) Pre-school non formal education
d) All of these
71. South Africa is leading exporter of which mineral?
a) Copper
b) Diamond
c) Silver
d) Gold
72. The word 'sustainable development' came into existence in the year.
a) 1992
b) 1978
c) 1980
d) 1987
73. The other word of landscaping is
a) Reduction
b) Restoration
c) Removing topsoil
d) Restore
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79. Which of the following is biodegradable?
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b) Domestic sewage
c) Detergents
d) a and c

80. Blaring sounds known to cause
 a) Mental distress
 b) High cholesterol
 c) Neurological problems
 d) All of these
81. In geosynchronous orbit altitude of the satellite is about
 a) 36,000 kms
 b) 10,000 kms
 c) 50,000 kms
 d) None of these
82. The Air (Prevention and Control of Pollution) Act was enacted in the year.
 a) 1987
 b) 1981
 c) 1991
 d) 1988
83. Kudremukh Iron ore mine, Karnataka was closed due to
 a) River pollution and threat to biodiversity
 b) Land encroachment
 c) Radioactive hazards
 d) Serious health hazard
84. On the eve of Gandhi Jayanthi which andolan was launched by our Honorable Prime Minister
 a) Swadeshi
 b) Sarvashikshana Abhiyana
 c) Suvarnagrama
 d) Swach Bharath
85. An international agreement signed in the year 1987, to protect stratospheric ozone is known as
 a) Montreal protocol
 b) Kyoto protocol
 c) Earth summit
 d) None of these
86. The explosion of First Atomic Bomb was done in Hiroshima and Nagasaki in
 a) 1946
 b) 1986
 c) 1945
 d) 1947
87. A dangerous pesticide which has been reported to cause physical deformities to people of Kerala and Karnataka states
 a) Endosulfan
 b) Fluorides
 c) DDT
 d) Dioxygene
88. Visible portion of EMR ranges between
 a) 0.4 - 0.76 μm
 b) 10.5 - 12.5 μm
 c) 8.0 - 14.0 μm
 d) None of these
89. Data representation in Raster data is by
 a) pixel
 b) points, lines and polygon
 c) latitude and longitude
 d) none of these
90. In water treatment, alum is used for
 a) softening
 b) coagulation
 c) filtration
 d) disinfection
91. Which of the following element make e-waste hazardous in nature?
 a) Land
 b) Glass
 c) Plastic
 d) Iron
92. What is the hazardous pollutant released from batteries?
 a) Arsenic
 b) Barium
 c) Cobalt
 d) Cadmium
93. What is the term used for reuse of sewage sludge?
 a) Compost
 b) Solids
 c) Biosolids
 d) Sludge

94. Reduction in brightness of the famous Taj Mahal is due to
a) Global warming
b) Air pollution
c) Ozone depletion
d) Afforestation
95. E.I.A. can be expanded as
a) Environment and Industrial Act
b) Environment of Impact Activities
c) Environmental Impact Assessment
d) Environmentally Important Activity
96. Organic Farming is
a) Farming without using pesticides and chemical fertilizer
b) Enhances biodiversity
c) Promotes soil biological activity
d) All of these
97. Bio-remediation means the removal of contaminants from
a) Soil
b) Wastewater
c) Groundwater
d) Both soil and ground water
98. Plants use _____ gas for photosynthesis.
a) Oxygen
b) Methane
c) Nitrogen
d) Carbon dioxide
99. What is the maximum allowable concentration of fluorides in drinking water?
a) 1.0 mg/l
b) 1.25 mg/l
c) 1.50 mg/l
d) 1.75 mg/l
100. Forest rich area in Karnataka is found in _____.
a) Western Ghats
b) Bandipur
c) Nagarhole
d) Mangalore

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Question Paper Version : D

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
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- The term used for reuse of sewage sludge?
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- The loss in brightness of the famous Taj Mahal is due to
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 b) Air pollution
 c) Acid depletion
 d) Afforestation
- The Environment Protection Act can be expanded as
 a) Environment Protection and Industrial Act
 b) Environment of India
 c) Environmental Impact Assessment
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- Organic farming is
 a) Farming without using pesticides and chemical fertilizer
 b) Farming without using pesticides and chemical fertilizer
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- Organic farming
 a) Increases biodiversity
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 a) The area in Karnataka is found in
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- Organic farming
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- monoxide
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a) temperature
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c) wind
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a) 2
b) 5
c) 3
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a) High density of population
b) Only endangered plants
c) High density of population and plants
d) High density of bio-diversity
- _____ % of the earth's surface is covered by water.
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- _____ is an event arising out of
a) hazard event
b) causes of hazard event
c) disaster event
d) all of these
- The scientific study of earthquake is called
a) seismology
b) seismology
c) seismology
d) none of these
- "Mad Hatter Disease" is caused due to
a) Arsenic
b) Arsenic
c) Mercury
d) Mercury
- _____ is an eco-friendly fuel for automobiles is
a) Diesel
b) Diesel
c) CNG
d) CNG
- _____ explosion will cause
a) Stress on ecosystem
b) Stress on ecosystem
c) Stress on ecosystem
d) None of these
- Which of the following is having high population density?
a) China
b) China
c) USA
d) USA

78. Which of the following is an air pollutant?
a) Carbon dioxide
b) Oxygen
c) Nitrogen
d) Particulate matter
 79. Cyto toxic and expired drugs are disposed of by
a) damping
b) autoclave
c) incineration
d) chemical disinfection
 80. The colour code of plastic bag for disposing of microbial laboratory culture waste.
a) Black
b) Red
c) Blue
d) White
 81. Among the fresh water available in the earth the percentage of surface water is about
a) 50%
b) 10%
c) 5%
d) less than 1%
 82. Hepatitis is caused by
a) Protozoa
b) Virus
c) Bacteria
d) Fungus
 83. In India groundwater resources are rich in
a) Plains of river Kaveri and Krishna
b) The Deccan plateau
c) The Gangetic plains
d) The plains of Netravati and Kapila
 84. The required iron content in drinking water as specified by BIS is
a) 300 mg/l
b) 30 mg/l
c) 3 mg/l
d) 0.30 mg/l
 85. Molasses from sugar industry is used to generate
a) Biodiesel
b) Hydrogen
c) Bioethanol
d) Biomethanol
 86. Wind Farms are located in
a) River basin
b) Plain area
c) Hilly area
d) Valley area
 87. Biomass consists of
a) Lignin
b) Hemi cellulose
c) Cellulose
d) All of these
 88. Natural gas contains
a) Carbon dioxide
b) Hydrogen
c) Methane
d) Nitrogen
 89. Anti tobacco day is mentioned on
a) 31st May
b) 30st June
c) 31st July
d) 31st August
 90. Population explosion will cause
a) Socio-Economic Problems
b) Food Scarcity
c) Energy crises
d) All of these
 91. GIS stands for
a) Geostationary Interact Sector
b) Geographical Information System
c) Geotechnical Information Society
d) Geothermal Investigation Site

92. LPG is a mixture of
a) N_2 and H_2S b) CO_2 and N_2
c) Propane and butanes d) Methane and CO_2
93. The Tiger Conservation Project was started in
a) 1973 b) 1975 c) 1981 d) 2000
94. The leader of "Chipko Movement" is
a) Sunderlal Bahuguna b) Medha Patkar
c) Vandana Shiva d) Mahatma Gandhi
95. Which of the following is the source of Fly-ash?
a) Vehicular exhaust b) Sewage
c) Thermal power plant d) All of these
96. The permissible range of pH for drinking water as per the Indian Standard is
a) 6 to 9 b) 6.5 to 8.5
c) 6 to 8.5 d) 6.5 to 7.5
97. Water logging is a phenomena in which
a) Water patterns are rotated
b) Soil root zone becomes saturated due to over irrigation
c) Erosion of soil
d) Soil degradation
98. Carbon content is higher in
a) Living matter b) Soil
c) Water d) Atmosphere
99. Springs means
a) Surface water b) Atmosphere water
c) Both (a) and (b) d) Ground water
100. Bio-remediation means deliberately introducing micro organisms to break pollutants.
a) Soil b) Waste water
c) Ground water d) Both soil and groundwater

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CBCS SCHEME

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18EC71

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023 Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Physical Topology? With a neat diagram, explain the various types of physical topologies available in computer networks. (10 Marks)
- b. With a neat diagram, explain the significance of layers in TCP/IP protocol suite. (10 Marks)

OR

- 2 a. Explain LAN and WAN with the help of neat diagrams. (06 Marks)
- b. With a neat diagram, explain the five components of Data Communication. (06 Marks)
- c. Explain encapsulation and decapsulation in TCP/IP model with the help of a neat diagram. (08 Marks)

Module-2

- 3 a. What is an ARP? Explain the operation of ARP and its packet format with suitable diagrams. (10 Marks)
- b. Explain stop and wait protocol with a neat FSM diagram. Also explain how sequence and acknowledge numbers prevent duplication of frames with necessary diagrams. (10 Marks)

OR

- 4 a. A slotted ALOHA network transmits 200 bit frames using a shared channel with a 200 kbps bandwidth. Find the throughput if the system produces
(i) 1000 frames per second (ii) 500 frames per second (iii) 250 frames per second? (06 Marks)
- b. Explain CSMA/CA protocol with a flow diagram. (08 Marks)
- c. Explain the Ethernet Frame format of standard Ethernet. (06 Marks)

Module-3

- 5 a. Explain with a neat diagram, the virtual circuit packet switched network and its various phases of operation. (10 Marks)
- b. With a neat diagram explain IPv4 Datagram format. (10 Marks)

OR

- 6 a. Explain with an example, the Distance Vector Routing algorithm. (10 Marks)
- b. Explain with an example, Link State Routing and also apply Dijkstra algorithm to find least cost path tree. (10 Marks)

Module-4

- 7 a. Explain connectionless and connection oriented protocols in transport layer. (10 Marks)
- b. With a neat diagram, explain state transition diagram of TCP. (10 Marks)

OR

- 8 a. Explain Go-Back-N protocol along with sliding window diagrams. (10 Marks)
b. Explain TCP connection establishment using three way hand shaking. (10 Marks)

Module-5

- 9 a. Explain World Wide Web and Web documents with necessary diagrams. (10 Marks)
b. Explain the Architecture of Electronic mail with a neat diagram. (10 Marks)

OR

- 10 a. Explain with an example, the working of Hyper Text Transfer Protocol. (10 Marks)
b. What is Name-address resolution? With a neat diagram, explain the various types of resolution that are available. (10 Marks)

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CBCS SCHEME

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18EC71

Seventh Semester B.E. Degree Examination, June/July 2023 Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define what is data communication. Write and explain components of a data communication system. (10 Marks)
- b. Mention and explain different formats used to represent information in data communication. (10 Marks)

OR

- 2 a. Explain different data flow techniques used for communication between two devices. (10 Marks)
- b. With layer diagram, explain the responsibility of each layer in OSI model. (10 Marks)

Module-2

- 3 a. Explain the need for protocol layering. What are its advantages. (10 Marks)
- b. With neat sketch, explain encapsulation and decapsulation. (10 Marks)

OR

- 4 a. Define and explain the following terms :
 - i) Framing
 - ii) Flow control
 - iii) Error control.(09 Marks)
- b. What is piggybacking? Explain concept of piggybacking with neat diagram. (11 Marks)

Module-3

- 5 a. What is Controlled Access? Explain different control access methods. (10 Marks)
- b. A slotted ALOHA networks transmits 200 bit frame on a shared channel of 200 Kbps. Find the through put, if the system produces
 - i) 1000 frames per second
 - ii) 500 frames per second
 - iii) 250 frames per second.(10 Marks)

OR

- 6 a. Explain different forwarding techniques used to forward the packets from source to destination. (10 Marks)
- b. What is address space? Explain different classes IP addresses. (10 Marks)

Module-4

- 7 a. Write a note on security of IPV4 datagrams. (10 Marks)
b. With suitable diagram explain distance vector routing. (10 Marks)

OR

- 8 a. Explain the services provided by transport layer to the upper layer. (10 Marks)
b. Explain stop and wait ARQ protocol. (10 Marks)

Module-5

- 9 a. Explain TCP UDP datagram. (12 Marks)
b. Explain TCP connection establishment and connection termination. (08 Marks)

OR

- 10 a. Explain how application layer protocol interacts with End – user applications. (10 Marks)
b. Explain working of any one applications layer protocol. (10 Marks)

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Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

VLSI Design

Time: 3 hrs.

Max. Marks: 100

*Note: Answer any FIVE full questions, choosing ONE full question from each module.***Module-1**

- 1 a. With neat graph define Moore's law. Explain the history of integrated structures. (06 Marks)
- b. Realize CMOS logic structure for the Boolean expressions :
 i) $y = \overline{(a \cdot b)} + (c \cdot d)$ ii) $y = a \cdot (b + c)$. (06 Marks)
- c. With neat diagrams, explain 3 regions of operations of nMOS transistor. (08 Marks)

OR

- 2 a. List the any three non ideal features of transistors. Explain each in detail. (06 Marks)
- b. Draw the diagram of general logic gate structure. Explain 2-input CMOS NAND gate functioning using truth table. (06 Marks)
- c. Draw schematic diagram of CMOS inverter. Explain the graphical derivation of CMOS inverter DC characteristics. (08 Marks)

Module-2

- 3 a. With neat diagrams, explain the complete CMOS fabrication process. (12 Marks)
- b. Using relevant equations explain full scaling (constant field scaling) applied to
 i) Channel length ii) Channel depth iii) Oxide thickness iv) Junction depth v) Supply
 vi) Threshold voltage vii) Doping densities N_A , N_D . (08 Marks)

OR

- 4 a. Write a short note on timing analyzer. (06 Marks)
- b. With neat diagrams, explain the lumped representation of parasitic MOSFET capacitances. (08 Marks)
- c. Draw and explain layout rules for transistors. (06 Marks)

Module-3

- 5 a. Explain various stages of timing optimization in VLSI design. (08 Marks)
- b. With equations explain the calculation of inverter delay. (06 Marks)
- c. Estimate the propagation delay t_{pd} for unit inverter driving 'm' identical unit inverters using Elmore delay. (06 Marks)

OR

- 6 a. Draw the diagram of photo masking with a negative resist and explain. (08 Marks)
- b. What is logical effort? Explain HI-Skew inverter construction by down sizing of nMOS transistor. (06 Marks)
- c. Explain pseudo nMOS inverter with schematic diagram and DC transfer characteristics. (06 Marks)

Module-4

- 7 a. Draw and explain the functioning of pulse generators. (08 Marks)
b. Explain the working of resettable flip-flops and latches. (12 Marks)

OR

- 8 a. Draw and explain the features of C²MOS latch. (08 Marks)
b. With neat circuit diagrams, explain 4 transparent latches. Write the advantage and disadvantage of each. (12 Marks)

Module-5

- 9 a. Draw the diagram of 4 bit × 4 bit NOR based ROM array, explain the functioning. (08 Marks)
b. What is static RAM? With neat diagram explain any 3 static RAM circuits. (12 Marks)

OR

- 10 a. Write a short note on design for testability. (06 Marks)
b. Explain manufacturing test principles in detail. (06 Marks)
c. Explain the logic verification principles. (08 Marks)

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18EC72

Seventh Semester B.E. Degree Examination, June/July 2023 VLSI Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Implement a 4 : 1 multiplexer using :
 - i) Transmission gate
 - ii) Tristate inverters

(08 Marks)
- b. Realize CMOS compound gate for the function : $Y = \overline{D + A(B + C)}$.

(04 Marks)
- c. With necessary circuit diagram and timing diagram explain the operation of positive edge triggered D flip-flop.

(08 Marks)

OR

- 2 a. Draw the circuit diagram of a CMOS inverter and its DC transfer characteristics. Explain various region of operation and indicate the voltage levels. Derive the equation for switching threshold.

(10 Marks)
- b. Derive the equation for drain current of a MOSFET in non-saturated and saturated region of operation.

(06 Marks)
- c. Explain the following non-ideal effects of a MOSFET –channel length modulation mobility degradation.

(04 Marks)

Module-2

- 3 a. With necessary diagrams explain CMOS n-well fabrication process.

(12 Marks)
- b. Draw the layout of $Y = ABC + D$ and estimate the area.

(08 Marks)

OR

- 4 a. With necessary diagrams explain lambda based design rules for wires, contact cuts and transistors.

(08 Marks)
- b. Explain MOSFET capacitances in three different regions of operation with necessary diagrams and equations.

(06 Marks)
- c. What is Scaling? Compute drain current, power, current density, power density, C_{ox} for constant field scaling.

(06 Marks)

Module-3

- 5 a. Explain the RC delay model to compute the delay of the logic circuit. Also calculate the delay of unit size inverter driving another unit size inverter.

(08 Marks)
- b. With necessary circuit example explain :
 - i) Pseudo nMOS
 - ii) Ganged CMOS.

(06 Marks)
- c. Explain the following CMOS optimization techniques with necessary examples :
 - i) Input ordering
 - ii) Asymmetric gates.

(06 Marks)

OR

- 6 a. Analyze the three input NAND gate using Elmore's delay and compute the falling and rising propagation delays if the output is loaded with 'h' identical gates. (08 Marks)
- b. Compute and compare the logical effort and parasitic delay of the following gates with the help of schematic diagram :
 i) 2 input NOR gate
 ii) Input NAND gate. (06 Marks)
- c. Explain Cascade voltage switch logic (CVSL) implement two input OR/NOR gate using CVSL. (06 Marks)

Module-4

- 7 a. Compute the output voltage V_{out} in the following pass transistor circuits. Assume $V_{tn} = 0.7V$.

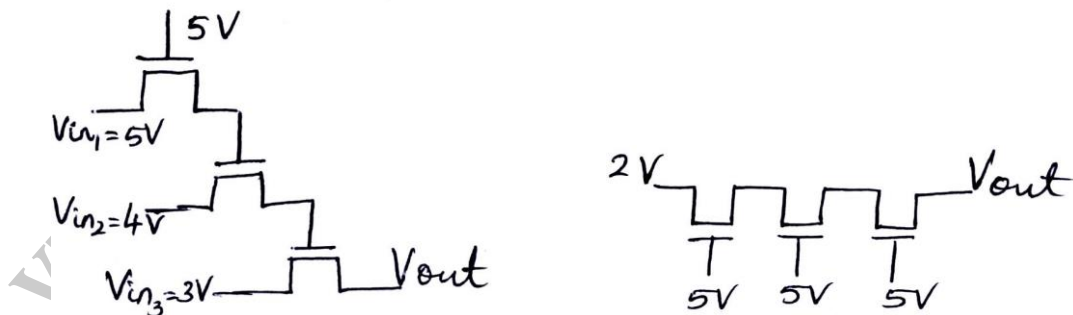


Fig.Q7(a)

- b. With necessary diagrams and equations explain charge storage and charge leakage in dynamic logic. (08 Marks)
- c. With necessary circuit diagrams explain resettable latches with :
 i) synchronous reset ii) asynchronous reset. (06 Marks)

OR

- 8 a. Explain dynamic logic with an example. Also explain the advantage and limitations of dynamic logic. (08 Marks)
- b. With necessary circuit diagram explain 3 bit dynamic shift register with enhancement load (radio less). (08 Marks)
- c. Explain dynamic synchronous CMOS transmission gate logic with necessary diagrams. (04 Marks)

Module-5

- 9 a. With necessary circuit diagram explain the operation of four transistor DRAM cell. (06 Marks)
- b. Explain the terms : i) controllability ii) observability iii) repeatability iv) survivability. (08 Marks)
- c. Explain full CMOS SRAM cell with necessary circuit topology. (06 Marks)

OR

- 10 a. Explain CMOS bridging fault with necessary example. (06 Marks)
- b. What is a fault model? Explain stuck at fault model with examples. (08 Marks)
- c. Draw the circuit of 3 bit BILBO register and explain. (06 Marks)

CBCS SCHEME

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18EC733

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a block diagram, explain the fundamental steps involved in Digital Image Processing. (10 Marks)
- b. Explain the process of image acquisition using a single sensor. (06 Marks)
- c. A common measure of transmission for digital data is baud rate (number of bits/Sec.). Each packet consists of byte, a start bit and a stop bit. How many minutes would it take to transmit a 1024×1024 image with 256 gray levels using : i) 33.6K baud modem ii) 200K baud modem. (04 Marks)

OR

- 2 a. Explain the components of a general purpose image processing system, with block diagram. (08 Marks)
- b. Explain with neat diagram, how image is acquired using Linear Sensor Strip. (06 Marks)
- c. Explain the importance of brightness adaptation and discrimination in image processing. (06 Marks)

Module-2

- 3 a. Explain the process of image sampling and quantization with an example. (07 Marks)
- b. Perform histogram equalization of a 10×10 , 3 – bit image having following distribution. Plot the input and output histogram.

Gray level	0	1	2	3	4	5	6	7
Number of pixels	50	25	15	10	0	0	0	0

(07 Marks)

- c. Explain image sharpening in spatial domain using second order Laplacian derivative. (06 Marks)

OR

- 4 a. Explain the concept of : i) Gray level slicing ii) Bit plane slicing. (08 Marks)
- b. Compute the lengths of the shortest 4-, 8- and m-path between p and q in the image segment shown in Fig.Q4(b) by considering $V = \{1, 2\}$.

	3	1	2	1	(q)
	2	2	0	2	
	1	2	1	1	
(p)	1	0	1	2	

Fig.Q4(b)

(06 Marks)

- c. Explain image negative. Determine the negative of a 4 bit image shown in Fig.Q4(c). Plot the histogram of the original image and its negative.

0	12	14	11
5	10	4	8
4	9	7	5
2	15	8	1

Fig.Q4(c)

(06 Marks)

Module-3

- 5 a. Explain Homomorphic filters for image enhancement with necessary equations, block diagram and transfer function. (10 Marks)
- b. State the following properties of 2D – DFT.
i) Translation ii) Periodicity iii) Rotation iv) Convolution theorem. (04 Marks)
- c. Compare an Ideal Lowpass filter with a Butterworth Lowpass filter. (06 Marks)

OR

- 6 a. Explain with a block diagram, the basic steps for image filtering in frequency domain. (06 Marks)
- b. Explain sharpening of images in frequency domain using Ideal, Butterworth and Gaussian Highpass filter. (09 Marks)
- c. State and prove linearity property of 2D – DFT. (05 Marks)

Module-4

- 7 a. With necessary equations and graph, explain the following noise probability density functions :
i) Gaussian Noise ii) Exponential Noise iii) Uniform Noise. (06 Marks)
- b. What is inverse filtering? What are its disadvantages? How are they eliminated using Weiner filtering? (08 Marks)
- c. Explain Adaptive local noise reduction filter. (06 Marks)

OR

- 8 a. Assuming only the presence of noise in an image, explain the following mean filters.
i) Arithmetic mean filter
ii) Geometric mean filter. (06 Marks)
- b. Apply a 3×3 median filter for the marked pixels in the image shown in Fig.Q8(b) and write the result of the filtering.

10	11	12	13	14	15
11	255	45	30	23	32
12	23	22	0	22	34
13	12	24	22	12	12

Fig.Q8(b)

- c. Explain the estimation of degradation function using :
i) Observation
ii) Experimentation. (08 Marks)

Module-5

- 9 a. What is Pseudo color image processing? Explain intensity slicing as applied to pseudo color image processing. (06 Marks)
- b. With necessary expressions explain erosion and dilation. (08 Marks)
- c. Explain RGB color model. (06 Marks)

OR

- 10 a. With necessary expressions explain opening and closing operations. List the properties of opening and closing operations. (10 Marks)
- b. Write the equations for converting colors from
i) RGB to HSI ii) HSI to RGB. (10 Marks)

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18EC733

Seventh Semester B.E. Degree Examination, June/July 2023 Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain fundamental steps in digital image processing with block diagram. (10 Marks)
- b. Describe image acquisition using sensor strips. (10 Marks)

OR

- 2 a. What is Digital Image? Explain components of Digital Image Processing. (10 Marks)
- b. Explain Image Formation Model. (06 Marks)
- c. List the application of Digital Image Processing. (04 Marks)

Module-2

- 3 a. Explain the process of image sampling and quantization. (08 Marks)
- b. Image transmission is done in packets. A packet consists of a start bit, a byte of data, and a stop bit. Answer the following:
 - (i) How many minutes it would take to transmit 512×512 image with 256 gray levels at 300 baud rate?
 - (ii) What would be time at 9600 baud? (04 Marks)
- c. Consider the image segment shown:

4	2	3	② q
3	3	1	3
2	3	2	2
p ②	1	2	3

Compute the length of shortest 4, 8 and m paths between p and q where $v = \{1, 2\}$ (08 Marks)

OR

- 4 a. Apply histogram specification on image shown in Fig.Q4(a).

0	1	0	2
2	3	3	2
0	1	0	1
1	3	2	0

Fig.Q4(a)

Having $r_i = z_i = 0, 1, 2, 3$

$\Pr(r_i) = 0.25$ for $i = 0, 1, 2, 3$

$\Pr(z_0) = 0, \Pr(z_1) = 0.5, \Pr(z_2) = 0.5, \Pr(z_3) = 0$

(08 Marks)

- b. Explain the following transformation:

- (i) Image negative
- (ii) Gray level slicing
- (iii) Log transformation

(12 Marks)

Module-3

- 5 a. Illustrate homomorphic filtering in image enhancement. Derive suitable result. (10 Marks)
 b. Explain different smoothing frequency domain filter for image enhancement. (10 Marks)

OR

- 6 a. Explain with block diagram, basic steps for image filtering in frequency domain. (10 Marks)
 b. Explain any three sharpening frequency domain filters for image enhancement. (10 Marks)

Module-4

- 7 a. Explain how image degradation estimation done using :
 (i) Observation
 (ii) Mathematical modeling (10 Marks)
 b. Explain different types of order-statistic filters used in image restoration. (10 Marks)

OR

- 8 a. Discuss different types of mean filters used in image restoration. (10 Marks)
 b. Explain adaptive median filter used in image restoration. (10 Marks)

Module-5

- 9 a. Explain the following color models:
 (i) RGB color model
 (ii) HSI model (10 Marks)
 b. Given HSI = (356.99, 0.534, 0.3453), convert this into RGB model. (06 Marks)
 c. Define the following terms:
 (i) Hue
 (ii) Saturation
 (iii) Intensity
 (iv) Chromaticity (04 Marks)

OR

- 10 a. Explain opening and closing in image processing. (12 Marks)
 b. Draw block diagram of pseudocolor processing and explain it. (08 Marks)

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18EC741

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

IoT and Wireless Sensor Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define IoT. Write an equation representing actions and communication of data at various levels in IoT and explain. (05 Marks)
- b. Draw a neat diagram showing various blocks and subsystems of an IoT framework suggested by IBM and explain in brief. (10 Marks)
- c. What is the significance of MQTT protocol in IoT? Explain the functionalities of MQTT Broker. (05 Marks)

OR

- 2 a. With a neat diagram, explain IoT reference model suggested by CISCO. (08 Marks)
- b. Explain the following functionalities of gateway at data adaption layer.
- i) Transcoding
 - ii) Privacy
 - iii) Security
 - iv) Data enrichment and consolidation. (06 Marks)
- c. What is Constrained Application Protocol (COAP) in IoT? Briefly discuss features of COAP. (06 Marks)

Module-2

- 3 a. Write a neat diagram, showing how the four layers of TCP/IP suite generates data for the network and physical layer during internet communication. List various actions that occur during transmission of data. (08 Marks)
- b. Explain various classes of IP addresses with an example for each and also specify total number of bits required for Net ID and Host ID. (06 Marks)
- c. Briefly explain different types of cloud deployment models with suitable examples. (06 Marks)

OR

- 4 a. Draw a neat diagram showing various fields of IPv4 header and explain them in detail. (08 Marks)
- b. Draw an DODAG data flow graph for RPL network assuming routing nodes at Four ranks (levels) 0, 1, 2 and 3 consisting of 3, 6, 3 and 6 nodes respectively. Also list the characteristics of DODAG. (Destination oriented directed Acyclic Graph). (06 Marks)
- c. What are various cloud service models explain each of them with examples. (06 Marks)

Module-3

- 5 a. Explain how to program an embedded devices on Arduino platform using IDE? (05 Marks)
 b. Write a program for Arduino controlled traffic lights at a road junction with three traffic lights RED, YELLOW and GREEN to be controlled on each side of the four sides North, East. West and South clock wise path ways. (10 Marks)
 c. Explain how data is read from sensors and devices using :
 i) ADC analog input
 ii) The timers
 iii) Operating system
 iv) Software serial library
 v) The libraries. (05 Marks)

OR

- 6 a. List out the vulnerabilities identified by Open Web Application Security Project (OWASP) for IoT Applications/Services. (05 Marks)
 b. What are the five functional components required to secure an IoT architecture. Explain then in brief. (05 Marks)
 c. Explain layered attacker model with possible attacks and suggest the solutions for mitigating the attacks on the layers. (10 Marks)

Module-4

- 7 a. Describe the characteristics of wireless sensor networks. (08 Marks)
 b. What are sensor networks? With a neat diagram explain various hardware components of sensor node. (06 Marks)
 c. Write a neat diagram showing the structure of transceiver and also explain various operational states of transceiver. (06 Marks)

OR

- 8 a. Explain following programming paradigms.
 i) Concurrent programming
 ii) Process based programming
 iii) Event based programming. (06 Marks)
 b. What are the three types of mobility in wireless sensor networks? Explain. (06 Marks)
 c. Explain various optimization goals and figure of merits in WSN's. (08 Marks)

Module-5

- 9 a. Explain low duty cycle and wakeup concepts in WSN's. (06 Marks)
 b. With a neat diagram, explain working principle of CSMA protocol. (08 Marks)
 c. Explain basic working principle of S – MAC protocol. (06 Marks)

OR

- 10 a. What are the factors that influences the design of physical layer in WSN's? (04 Marks)
 b. Explain in detail working principle of LEACH Routing protocol in WSN. (08 Marks)
 c. Write short notes on :
 i) Energy Efficient Routing
 ii) Geographical Routing. (08 Marks)

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Seventh Semester B.E. Degree Examination, June/July 2023 IOT and Wireless Sensor Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain IBM IOT conceptual framework with equation. (10 Marks)
- b. Explain Co-ap SMS and CO-ap MQ used in message communication. (10 Marks)

OR

- 2 a. Explain modified OSI model with protocols used in each layer. (06Marks)
- b. Explain the functions of data management and consolidation gateway. (08 Marks)
- c. Explain the features of MQTT protocol. (06Marks)

Module-2

- 3 a. Explain the IPV₄ header format. (06 Marks)
- b. Explain four cloud service models. (06 Marks)
- c. With neat diagram, explain the features of 6LOWPAN adaptation layer protocol for IEEE802.15.4 for network device. (08 Marks)

OR

- 4 a. Explain four layer data stack generated during Internet communication. (08 Marks)
- b. Explain the features of IPV6. (04 Marks)
- c. Explain the features of HTTP protocol. (08 Marks)

Module-3

- 5 a. Explain five levels of software development for the applications and service in IOT. (07Marks)
- b. Discuss about programming Embedded. Device Arduino platform using IDE. (07 Marks)
- c. Define Vulnerability and mention top ten vulnerabilities in IOT identified by OWASP. (06 Marks)

OR

- 6 a. Write a Arduino program to control the traffic light. (06 Marks)
- b. Explain IOT security Tomography and layered attack model. (06 Marks)
- c. What are the security requirements in IOT architecture and threat analysis using Microsoft threat model. (08 Marks)

Module-4

- 7 a. Explain the hardware components of single node architecture. (06 Marks)
b. Explain the energy consumed in sensor node fan operation states with different power consumption. (06 Marks)
c. Explain the optimization goal and figure of merit in WSN. (08 Marks)

OR

- 8 a. What are the desired characteristics to overcome the challenges in WS wire Sensor network? (08 Marks)
b. What are the different types of mobility in WSN? (06 Marks)
c. Explain the transceiver modules used in single node architecture. (06 Marks)

Module-5

- 9 a. Explain SMAC protocol. (10 Marks)
b. With neat schematic diagram, explain CSMA protocol. (10 Marks)

OR

- 10 a. Explain low duty cycle and wake up concepts in WSN. (06 Marks)
b. Explain LEACH protocol. (07 Marks)
c. Explain any two considerations in energy efficient unicast routing with example. (07 Marks)

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Third Semester B.E. Degree Examination, Jan./Feb. 2023
Transform Calculus, Fourier Series and Numerical
Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Find the Laplace transform of $te^{2t} \frac{2 \sin 3t}{t}$. (06 Marks)

- b. Given that $f(t) = \begin{cases} B, & 0 < t < \frac{\gamma}{2} \\ E, & \frac{\gamma}{2} < t < a \end{cases}$

where $f(t+a) \equiv f(t)$ show that $L\{f(t)\} = \frac{E}{s} \tan h\left(\frac{\pi s}{4}\right)$. (07 Marks)

- c. Using convolution theorem obtain the inverse Laplace transform of the following function:

$$\frac{(s-1)(s^2+1)}{(s-1)(s^2+1)}$$

(07 Marks)

OR

- 2 a. Find the inverse Laplace transform of:

$$\frac{s+5}{s^2-6s+13}$$

(06 Marks)

- b. Express the following function in terms of unit step function and hence find their Laplace transform.

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ t^2, & 1 < t \leq 2 \\ 0, & t > 2. \end{cases}$$

(07 Marks)

- c. Solve the following initial value problem by using Laplace transform:

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 0$$

(07 Marks)

Module-2

- 3 a. Obtain Fourier series of $f(x) = \frac{\pi-x}{2}$ in $0 < x < 2\pi$. Hence deduce that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$$

(06 Marks)

- b. Find a cosine Fourier series for $f(x) = (x-1)^2$, $0 \leq x \leq 1$.

(07 Marks)

- c. Obtain the Fourier series of y upto the first harmonic for the following values.

x°	45	90	135	180	225	270	315	360
y	4.0	3.8	2.4	2.0	-1.5	0	2.8	3.4

(07 Marks)

1 of 3

OR

- 4 a. Obtain Fourier series for

$$f(x) = \begin{cases} \pi x & \text{in } 0 \leq x \leq 1 \\ \pi(2-x) & \text{in } 1 \leq x \leq 2 \end{cases}$$

(06 Marks)

- b. Obtain the sine half range series for the function:

$$f(x) = 1 - \left(\frac{x}{\pi}\right) \ln \quad 0 \leq x \leq \pi.$$

(07 Marks)

- c. The following values of y and x are given. Find Fourier series of upto first harmonics.

x	0	2	4	6	8	10	12
y	9.0	18.2	24.4	27.8	27.5	22.0	9.0

(07 Marks)

Module-3

- 5 a. If $f(x) = \begin{cases} 1-x^2, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$. Find Fourier transform of $f(x)$ and hence find the value of

$$\int_0^\infty \frac{x \cos x - \sin x}{x^3} dx.$$

(06 Marks)

- b. Find the Fourier sine transform of $f(x) = e^{-|x|}$ and hence evaluate

$$\int_0^\infty \frac{x \sin mx}{1+x^2} dx, \quad m > 0.$$

(07 Marks)

- c. Solve by using Z-Transforms $U_{n+2} + 2U_{n+1} + U_n = n$ with $U_0 = 0 = U_1$.

(07 Marks)

- 6 a. Obtain the Fourier cosine transform of the function:

$$f(x) = \begin{cases} 4x, & 0 < x < 1 \\ 4-x, & 1 < x \leq 4 \\ 0, & x > 4. \end{cases}$$

(06 Marks)

- b. Obtain the Z-transform of $\cos n\theta$ and $\sin n\theta$.

(07 Marks)

- c. Compute the inverse Z-transform of $\frac{3z^2-2z}{(5z-1)(5z+2)}$.

(07 Marks)

Module-4

- 7 a. Classify the following partial differential equations:

$$i) \quad x^2 u_{xx} + (1-y^2) u_{yy} = 0, \quad -\infty < x < \infty, -1 < y < 1$$

$$ii) \quad (1+x^2) u_{xx} + (5+2x^2) u_x + (4+x^2) u_{yy} = 0$$

$$iii) \quad (x+1) u_{xx} - 2(x+2) u_{xy} + (x+3) u_{yy} = 0.$$

(10 Marks)

- b. Solve $u_t = u_{xx}$ subject to the conditions $u(0, t) = 0 = u(1, t)$ and $u(x, 0) = \sin \pi x$ by taking $h = 0.2$ for 5 levels. Further write down the following values from the table

$$i) \quad u(0.4, 0.08) \\ ii) \quad u(0.6m, 0.06).$$

(10 Marks)

2 of 3

OR

8. a. Solve the elliptic equation $u_{xx} + u_{yy} = 0$ for the following square Mesh with boundary values as shown. Find the iterative values of u_4 (to 9) to the nearest integer.



Fig. Q8(a)

- b. Solve $25u_x = u_t$ at the pivotal points given $u(0, t) = 0 = u(5, t)$, $u(x, 0) = 0$ and $u(x, 0) = \begin{cases} 20x, & 0 \leq x \leq 1 \\ 5(5-x), & 1 \leq x \leq 5 \end{cases}$ by taking $h = 1$ compute $u(x, t)$ for $0 \leq t \leq 1$. (10 Marks)

Module-5

9. a. Given $y'' - xy' - y = 0$ with the initial conditions $y(0) = 1, y'(0) = 0$ compute $y(0.2)$ using fourth order Runge-Kutta method. (06 Marks)
 b. Derive the Euler's equation. (07 Marks)
 c. Find the extremal of the functional. (07 Marks)

$$\int_{x_1}^{x_2} (y^2 + y'^2 + 2ye^x) dx.$$

OR

10. a. Obtain the solution of the equation $2 \frac{d^2 y}{dx^2} = 4x + \frac{dy}{dx}$ by computing the value of $y(1.4)$ by applying Milne's method using following data :

x	1	1.1	1.2	1.3
y	2	2.2156	2.4649	2.7514
y'	2	2.3178	2.6725	3.0657

(06 Marks)

- b. Find the curve for which the functional $\int_0^1 [y'^2 + 12xy] dx$ with $y(0) = 0$ and $y(1) = 1$ can be determined. (07 Marks)
 c. Prove that the shortest distance between two points in a plane is straight line. (07 Marks)

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Third Semester B.E. Degree Examination, Jan./Feb. 2023

Digital System Design using Verilog

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are combinational circuits? Give example. Explain combinational circuit with block diagram. (04 Marks)
- b. Define canonical form representation and solve the following equation using canonical form
 - i) $P = f(a,b,c) = ab' + ac' + bc$
 - ii) $G = f(w,x,y,z) = w'x + yz'$. (08 Marks)
- c. Simplify the following Boolean function using K – Map
 - i) $D = f(x, y, z) = \sum m(0, 2, 4, 6)$
 - ii) $K = f(a, b, c) = \sum m(1, 2, 3, 6, 7)$. (08 Marks)

OR

- 2 a. Define K-Map solve the following expression using K – Map.
 - i) $K = f(w, x, y, z) = \sum m(0, 1, 4, 5, 9, 11, 13, 15)$
 - ii) $D = f(a, b, c, d) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$. (10 Marks)
- b. Define Quine-McClusky method and solve the following Boolean expression using Quine-McClusky method.
 - i) $D = f(a, b, c, d) = \sum m(0, 1, 2, 3, 6, 7, 8, 9, 14, 15)$
 - ii) $K = f(w, x, y, z) = \sum m(1, 3, 13, 15) + \sum d(8, 9, 10, 11)$. (10 Marks)

Module-2

- 3 a. Explain binary Adders with K-map and logical representation of equations for SUM and CARRY. (06 Marks)
- b. Explain carry look ahead Adder with General and Sigma block. (06 Marks)
- c. Explain working of decimal adder with neat block diagram (take example of BCD addition). (08 Marks)

OR

- 4 a. What are comparator circuits? Explain 2-bit magnitude comparators. (08 Marks)
- b. Realize the Boolean expression using 3 : 8 decoder and two OR gates
 - i) $f_1(x_2, x_1, x_0) = \sum m(1, 2, 4, 5)$
 - ii) $f_2(x_2, x_1, x_0) = \sum m(1, 5, 7)$. (06 Marks)
- c. Implement $D = (w, x, y, z) = \sum m(0, 1, 2, 4, 5, 7, 8, 9, 12, 13)$ using 8 : 1 MUX. (06 Marks)

Module-3

- 5 a. Write a note on Master Slave JK Flip-Flops with function table and timing diagram. (08 Marks)
- b. What are Edge Triggered Flip-Flops. Explain positive edge Triggered and negative edge Triggered Flip-Flops. (06 Marks)
- c. Write characteristic equation for : i) JK Flip-Flop ii) SR Flip-Flop. (06 Marks)

OR

- 6 a. Define Counters. Explain Binary Ripple counter with neat diagram. (08 Marks)
 b. What are Registers? Explain any two classification registers with neat block diagram. (06 Marks)
 c. Design synchronous MOD-6 counter using clocked JK Flip-Flops for sequences :
 0 – 2 – 3 – 6 – 5 – 1. (06 Marks)

Module-4

- 7 a. Define HDL and types of HDL. Give structure of verilog module with example. (06 Marks)
 b. Explain verilog logical operators with example. (06 Marks)
 c. i) Write a note on verilog Data type
 ii) Write verilog code for 8×1 MUX. (08 Marks)

OR

- 8 a. Give classification of Styles(Types) of description with example. (08 Marks)
 b. Write verilog code for Full Adder. (06 Marks)
 c. Write a note on Arithmetic and shift, Rotate relational operators with example. (06 Marks)

Module-5

- 9 a. Write a note on structure of Behavioural Description with example. (08 Marks)
 b. Write a note on Signal Assignment and Variable Assignment with example. (06 Marks)
 c. Write a note on sequential statement with example. (06 Marks)

OR

- 10 a. Write a verilog code for 2×1 MUX using if ELSE STATEMENT. (06 Marks)
 b. Explain structural description with example. (08 Marks)
 c. Explain structural description of 3-bit Ripple Carry Adder. (06 Marks)

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21EC33

Third Semester B.E. Degree Examination, Jan./Feb. 2023

Basic Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain vector spaces and its necessary axioms. And also explain four fundamental subspaces with example. (08 Marks)
- b. Write the vector $V = (1, 3, 9)$ as a linear combination of the vectors $u_1 = (2, 1, 3)$, $u_2 = (1, -1, 1)$ and $u_3 = (3, 1, 5)$ and thereby show that the system is consistent. (08 Marks)
- c. Let $I: V_1(\mathbb{R}) \rightarrow V_2(\mathbb{R})$ be a mapping $f(x) = (3x, 5x)$ show that 'f' is linear transformation. (04 Marks)

OR

- 2 a. Let 'w' be the subspace of $\mathbb{R}^5$ spanned by
 $x_1 = (1, 2, -1, 3, 4)$, $x_2 = (2, 4, -2, 6, 8)$, $x_3 = (1, 3, 2, 2, 6)$,
 $x_4 = (1, 4, 5, 1, 8)$, $x_5 = (2, 7, 3, 3, 9)$.
 Find a subset of vectors which forms a basis of 'w'. (06 Marks)
- b. Solve $Ax = b$ by least square and find $P = A \hat{x}$ if
 $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}_{3 \times 2}$ and $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}_{3 \times 1}$. Also, write a program to solve linear equation $Ax = b$. (07 Marks)
- c. Apply Gram – Schmidth process to the vectors $V_1(1, 1, 1)$, $V_2(1, -1, 2)$, $V_3(2, 1, 2)$ to obtain an orthonormal basis for $V_3(\mathbb{R})$ with standard inner product and thereby write a program for Gram – Schmidth process. (07 Marks)

Module-2

- 3 a. If $A = \begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$ find Eigen values and corresponding Eigen vector for matrix 'A' and diagonalize the matrix. (10 Marks)
- b. If $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$. Show that matrix 'A' is positive definite matrix using the following approaches :
 i) By finding its Eigen value
 ii) By finding its pivots. (10 Marks)

OR

- 4 a. Compute $A^T A$ and AA^T , find Eigen values and Eigen vectors, if $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}_{3 \times 2}$, thereby multiply $U \in V^T$ to recover matrix 'A'. Also write a program to find SVD. (12 Marks)

- b. Diagonalize the matrix A, if $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ by finding its eigen value and eigen vector. (08 Marks)

Module-3

- 5 a. Define signal and system and also explain basic discrete elementary signals with neat sketch and expressions. (04 Marks)
- b. A discrete time signal $x(n]$ is shown below Fig.Q5(b).

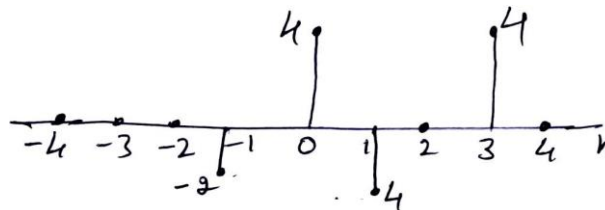


Fig.Q5(b)

Sketch :

- i) $2x(n-2)$
- ii) $3-x(n)$
- iii) $2x(-n)-4$.

(08 Marks)

- c. Sketch : $x(n) = \begin{cases} 1; & -1 \leq n \leq 3 \\ \frac{n}{2} & n = 4 \\ 0; & \text{otherwise} \end{cases}$ and $y(n) = \begin{cases} \frac{1}{2}n; & |n| \leq 4 \\ 0; & \text{otherwise} \end{cases}$

Also sketch $x(n+2)y(1-2n)$.

(08 Marks)

OR

- 6 a. For the following discrete time systems, determine whether the system is linear, time invariance, memoryless, causal and stable :

i) $y(n) = 2x(n) + \frac{1}{x(n-2)}$

ii) $y(n) = \ln(3 + |x(n)|)$

iii) $y(n) = \cos x(n)$

iv) $y(n) = r^n x(n); r > 1$.

(16 Marks)

- b. Write a program to generate exponential and triangular waveforms.

(04 Marks)

Module-4

- 7 a. Compute the discrete time convolution for the sequences $x_1(n)$ and $x_2(n)$ given below :
 $x_1(n) = \alpha^n u(n)$; $x_2(n) = \beta^n u(n)$. (08 Marks)
- b. Consider the input signal $x(n)$ and the impulse response $h(n)$ given below :
- $$x(n) = \begin{cases} 1; & 0 \leq n \leq 4 \\ 0; & \text{otherwise} \end{cases} \quad \text{and} \quad h(n) = \begin{cases} \alpha^n; & 0 \leq n \leq 6, \alpha > 1 \\ 0; & \text{otherwise} \end{cases}$$
- compute the output signal $y(n)$. (12 Marks)

OR

- 8 a. The following are the impulse responses of discrete time LTI systems. Determine whether each system is memoryless, causal and stable :
- i) $h(n) = e^{-n} \cos(n) \cdot u(n)$
- ii) $h(n) = (0.99)^n u(n+3)$
- iii) $h(n) = n\left(\frac{1}{2}\right)^n u(n)$. (10 Marks)
- b. Evaluate the step response of LTI system represented by the impulse response
 $h(n) = (-1)^n \{u(n+2) - u(n-3)\}$.
 Also write a program to compute the step response from the given impulse response. (10 Marks)

Module-5

- 9 a. Define Z-transform. Explain the properties of ROC. (06 Marks)
- b. Let $x(n) = \left(\frac{1}{2}\right)^{|n|}$.
- i) Sketch $x(n)$
- ii) Find $x(z)$ and sketch pole zero plot and ROC. (08 Marks)
- c. Find the Z-transform of $x(n) = \left(\frac{1}{2}\right)^n u(n) * \left(\frac{1}{3}\right)^n u(n)$. (06 Marks)

OR

- 10 a. Explain the properties of Z-transform with proof :
- i) Convolution
- ii) Initial value theorem
- iii) Final value theorem. (08 Marks)
- b. Determine the discrete time sequence $x(n)$ of the sequence using partial fraction expression:
- $$X(z) = \frac{-1 + 5z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}; \text{ROC: } |z| > 1$$
- (08 Marks)
- c. Write a program to find Z-transform of the sequence. (04 Marks)

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CBCS SCHEME

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21EC34

Third Semester B.E. Degree Examination, Jan./Feb. 2023

Analog Electronic Circuits

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the classical discrete circuit bias (voltage divider bias) method of BJT. (08 Marks)
- b. Explain the three biasing methods to bias MOS Amplifier circuits. (12 Marks)

OR

- 2 a. Explain the T equivalent circuit model of MOSFET. (08 Marks)
- b. Derive an expression for voltage gain of MOSFET with necessary waveforms. (06 Marks)
- c. Explain biasing a BJT using collector to base feedback resistor. (06 Marks)

Module-2

- 3 a. Explain the common source amplifier and derive the expression for voltage gain. (10 Marks)
- b. A transistor amplifier is fed with a signal source having an open circuit voltage V_{sig} of 10mV and an internal resistance r_{sig} of 100K Ω . The voltage V_i at the amplifier input and the output voltage V_o are measured both without and with load resistance $R_L = 10K\Omega$ connected to the amplifier output. The measured results are as follows :

	V_i (mv)	V_o (mv)
Without R_L	9	90
With R_L connected	8	70

Find all the amplifier parameters.

(10 Marks)

OR

- 4 a. With a neat diagram, explain the three frequency bands of MOSFET. (06 Marks)
- b. Explain the high frequency model of MOSFET. (06 Marks)
- c. Explain common source follower and derive the expression of voltage gain. (08 Marks)

Module-3

- 5 a. Explain the properties of negative feedback. (10 Marks)
- b. Explain the transformer coupled class – A power amplifier and show that efficiency is 50%. (10 Marks)

OR

- 6 a. Explain the circuit operation of class – B power amplifier and also explain the transfer characteristics. (08 Marks)
- b. Explain the Four basic feedback topologies of the amplifier. (12 Marks)

Module-4

- 7 a. Explain R and 2R resistor digital to analog (D/A) converter and also derive the expression of output voltage. (10 Marks)
- b. Explain the first order lowpass Butterworth filter with necessary voltage gain. (10 Marks)

OR

- 8 a. Explain the operation of monostable multivibrator. (10 Marks)
b. Explain the two types of Bandpass filters. (10 Marks)

Module-5

- 9 a. Explain the block diagram of power electronic system. (06 Marks)
b. List and explain the applications of power electronics. (06 Marks)
c. Explain the static anode – Cathode characteristics of SCR. (08 Marks)

OR

- 10 a. Explain the turn on methods of a Thyristor. (10 Marks)
b. Explain the construction and working of UJT. (10 Marks)

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Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023

CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- The idea of the Constitution of India was flashed for the first time by
 - Dr. B.R. Ambedkar
 - Dr. Rajendra Prasad
 - Mahatma Gandhiji
 - Jawaharlal Nehru
- The Indian Constitution came into force on
 - 26.11.1949
 - 26.01.1950
 - 15.08.1947
 - 26.12.1950
- The preamble of the Constitution of India has been amended so far
 - Four times
 - Thrice
 - Twice
 - Once
- Article 20 of the Indian Constitution refers to
 - Freedom of Speech
 - Right to Equality
 - Individual is Personal liberty
 - Abolition of Titles
- Who interprets the Indian Constitution?
 - Supreme Court
 - Parliament
 - President
 - Prime Minister
- Under the Constitution, the subjects of Administration have been divided into
 - Two lists
 - Four lists
 - Five lists
 - Three lists
- Which is the lengthiest Amendment to the Indian Constitution?
 - 46th
 - 44th
 - 42nd
 - 24th
- Which of these are the salient features of Indian Constitution?
 - Secularistic in Nature
 - Federal cum Unitary
 - Only 'a'
 - Both 'a' and 'b'
- The concept of Fundamental Duties are mentioned under :
 - Part II of the Constitution
 - Part III of the Constitution
 - Part IVA of the Constitution
 - Part V of the Constitution

10. Who among the following described the 'Preamble' as the "Horoscope of Indian Constitution"?
a) Mahatma Gandhiji b) J.J. Nehru c) K.M. Munshi d) Sardar Patel
11. The source of India's Sovereignty lies in the
a) People of India b) Supreme Court c) President d) Prime Minister
12. The Directive Principle of State Policy are
a) Justiciable b) Non Justiciable
c) Only some Directive Principles are Justiciable
d) None of these.
13. Which is not a Fundamental Right?
a) Right to Freedom b) Right to Constitutional remedies
c) Right to Property d) Right to Equality
14. Directive principles of State Policy have been described under the Article of :
a) 36 to 51 b) 1 to 11 c) 12 to 35 d) 19 to 27
15. Which one of the following is not a Fundamental duty?
a) To protect and improve natural Environment.
b) To develop Scientific temper
c) To abide by the Constitution.
d) To serve a Uniform civil code applicable to the entire Country.
16. Directive principles of the State policies were incorporated in the Constitution with a view to
a) Ensure a Democratic Government in the Country
b) Provide a strong Central Government
c) Establish Welfare state
d) Raise the Living Standards of the weaker sections of the Society.
17. The Constitution of India was formed by the Constituent Assembly under :
a) August offer of 1940 b) Cripps proposal of 1942
c) The Cabinet Mission Plan of 1946 d) Simon Commission of 1942
18. The Fundamental Rights granted by the Constitution of India to its Citizens cannot be suspended
a) Except by the order of the Supreme Court
b) Under any circumstances
c) Except by an order of the President during National Emergency
d) Except through an order of the President during War.
19. Right to Constitutional remedies is guaranteed under :
a) Article 21 b) Article 32 c) Article 30 d) Article 25
20. Which of these are Gandhian Principle?
a) Equal pay for equal work b) Prohibition of Cow slaughter
c) Respecting and Promoting Socialism and Democracy
d) Both b and c.
21. Who among the following is directly responsible to Parliament for all matters concerning the defense services of India?
a) President b) Cabinet Committee on Political Affairs
c) Prime Minister d) Defence Minister

22. Who among the following can attend meetings of the Union Cabinet?
 - a) President
 - b) Cabinet Ministers
 - c) Ministers of State
 - d) Deputy Ministers
23. Who acts as the President of India when neither the President nor the Vice President is available?
 - a) Senior most Governor of a State
 - b) Chief Justice of India
 - c) Speaker of Lok Sabha
 - d) Auditor General of India
24. Who among the following holds office during the pleasure of President?
 - a) Governor
 - b) Election Commissioner
 - c) Speaker
 - d) Prime Minister
25. If the President wants to resign from office, he may do so by writing to the
 - a) Chief Justice of India
 - b) Speaker of Lok Sabha
 - c) Prime Minister
 - d) Vice - President
26. The Union Council of Ministers is appointed by the
 - a) President of India according to his discretion
 - b) President of India on the advice of the Prime Minister
 - c) Prime Minister of India
 - d) Parliament
27. The Union Council of Ministers is collectively responsible to
 - a) Rajya Sabha
 - b) President
 - c) House of the People
 - d) Prime Minister
28. Who can initiate impeachment proceedings against the President of India.
 - a) Either House of Parliament
 - b) Any Vidhan Sabha
 - c) Only Lok Sabha
 - d) Rajya Sabha
29. The total number of members nominated by the President to the Parliament are :
 - a) 16
 - b) 14
 - c) 18
 - d) 21
30. The Chairman of the Planning Commission in India is the
 - a) President
 - b) Prime Minister
 - c) Minister of Planning
 - d) Finance Minister
31. The Election of the President is by a system of proportional representation by means of
 - a) Valid Transferable Vote
 - b) Transferable Vote
 - c) Single Transferable Vote
 - d) Legally Transferable Vote
32. When a Financial emergency is proclaimed
 - a) Union budget will not be presented
 - b) Salaries and allowances of any class of Employees may be reduced.
 - c) Repayment of Government debts will stop
 - d) Payment of Salaries to Public servants will be postponed.
33. This is not a ground to declare National emergency
 - a) War
 - b) Hung Legislature
 - c) Armed Rebellion
 - d) Failure of the Government
34. Who appoints the Election Commissioner
 - a) Prime Minister
 - b) Parliament
 - c) President
 - d) None of these

35. Proclamation of Emergency declared under Article 352 must be approved by the Parliament within
 a) 1 Year b) Six months c) 1 month d) 15 days
36. The control of the preparation of Electoral rolls for Parliament and State Legislature Elections rests with the
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37. Who appoints the Chairman for UPSC?
 a) Prime Minister b) Parliament c) Supreme Court d) President
38. Who is the Ex – officio Chairman of the Council of States?
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39. State emergency is declared on the written recommendation of :
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40. How many types of Emergencies have been mentioned in the Constitution of India?
 a) Four b) Three c) One d) Two
41. One of the characteristic of Profession is
 a) Monopoly b) Hard work c) Competition d) Money Minded
42. The term ethics is derived from
 a) Ethical in English b) Ethic in Latin c) Ethics in Greek d) French
43. Intellectual property is protected through
 a) The Patents , Trade mark and Copyrights b) Company Documentation
 c) Storage in Computers d) Security Personnel
44. Engineers first obligation is towards
 a) His Employer b) Public safety c) Government d) Clients
45. The codes of Ethics can be taken as guidelines by the Engineers to
 a) Resolve the conflicts b) Formulate the problems
 c) Overcome the work pressure d) Escape from the responsibility
46. Corrupt Professional Judgment leads to
 a) Integrity in R & D b) Conflicts of Interests c) Reliability d) None of these
47. Engineers will serve society better, if they are informed about
 a) Morality and code of conduct b) Technical standards
 c) Standards of Science d) Litigation process
48. Which of the following is not preserved as an Intellectual property?
 a) Government Regulations b) Patents
 c) Copyrights d) Trade Secrets
49. Which of these is a factor that affects ethical and unethical behavior?
 a) Diversity b) Ethical dilemma c) Team work d) Honesty
50. This is not the aim of studying Engineering Ethics
 a) Developing Ethics b) Natural Ethics
 c) Scientifically Developed Ethics d) Preventive Ethics

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Question Paper Version : B

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023

CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

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20. How many types of Emergencies have been mentioned in the Constitution of India?
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21. The source of India's Sovereignty lies in the
 - a) People of India
 - b) Supreme Court
 - c) President
 - d) Prime Minister
22. The Directive Principle of State Policy are
 - a) Justiciable
 - b) Non Justiciable
 - c) Only some Directive Principles are Justiciable
 - d) None of these.
23. Which is not a Fundamental Right?
 - a) Right to Freedom
 - b) Right to Constitutional remedies
 - c) Right to Property
 - d) Right to Equality
24. Directive principles of State Policy have been described under the Article of :
 - a) 36 to 51
 - b) 1 to 11
 - c) 12 to 35
 - d) 19 to 27

25. Which one of the following is not a Fundamental duty?
- To protect and improve natural Environment.
 - To develop Scientific temper
 - To abide by the Constitution.
 - To serve a Uniform civil code applicable to the entire Country.
26. Directive principles of the State policies were incorporated in the Constitution with a view to
- Ensure a Democratic Government in the Country
 - Provide a strong Central Government
 - Establish Welfare state
 - Raise the Living Standards of the weaker sections of the Society.
27. The Constitution of India was formed by the Constituent Assembly under :
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 - Cripps proposal of 1942
 - The Cabinet Mission Plan of 1946
 - Simon Commission of 1942
28. The Fundamental Rights granted by the Constitution of India to its Citizens cannot be suspended
- Except by the order of the Supreme Court
 - Under any circumstances
 - Except by an order of the President during National Emergency
 - Except through an order of the President during War.
29. Right to Constitutional remedies is guaranteed under :
- Article 21
 - Article 32
 - Article 30
 - Article 25
30. Which of these are Gandhian Principle?
- Equal pay for equal work
 - Prohibition of Cow slaughter
 - Respecting and Promoting Socialism and Democracy
 - Both b and c.
31. The idea of the Constitution of India was flashed for the first time by
- Dr. B.R. Ambedkar
 - Dr. Rajendra Prasad
 - Mahatma Gandhiji
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33. The preamble of the Constitution of India has been amended so far
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 - Individual is Personal liberty
 - Abolition of Titles
35. Who interprets the Indian Constitution?
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 - Parliament
 - President
 - Prime Minister
36. Under the Constitution, the subjects of Administration have been divided into
- Two lists
 - Four lists
 - Five lists
 - Three lists
37. Which is the lengthiest Amendment to the Indian Constitution?
- 46th
 - 44th
 - 42nd
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38. Which of these are the salient features of Indian Constitution?
a) Secularistic in Nature b) Federal cum Unitary
c) Only 'a' d) Both 'a' and 'b'
39. The concept of Fundamental Duties are mentioned under :
a) Part II of the Constitution b) Part III of the Constitution
c) Part IVA of the Constitution d) Part V of the Constitution
40. Who among the following described the "Preamble" as the "Horoscope of Indian Constitution".
a) Mahatma Gandhiji b) J.J. Nehru c) K.M. Munshi d) Sardar Patel
41. Who among the following is directly responsible to Parliament for all matters concerning the defense services of India?
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c) Prime Minister d) Defence Minister
42. Who among the following can attend meetings of the Union Cabinet?
a) President b) Cabinet Ministers
c) Ministers of State d) Deputy Ministers
43. Who acts as the President of India when neither the President nor the Vice President is available?
a) Senormost Governor of a State b) Chief Justice of India
c) Speaker of Lok Sabha d) Auditor General of India
44. Who among the following holds office during the pleasure of President?
a) Governor b) Election Commissioner
c) Speaker d) Prime Minister
45. If the President wants to resign from office, he may do so by writing to the
a) Chief Justice of India b) Speaker of Lok Sabha
c) Prime Minister d) Vice - President
46. The Union Council of Ministers is appointed by the
a) President of India according to his discretion
b) President of India on the advice of the Prime Minister
c) Prime Minister of India d) Parliament
47. The Union Council of Ministers is collectively responsible to
a) Rajya Sabha b) President
c) House of the People d) Prime Minister
48. Who can initiate impeachment proceedings against the President of India.
a) Either House of Parliament b) Any Vidhan Sabha
c) Only Lok Sabha d) Rajya Sabha
49. The total number of members nominated by the President to the Parliament are :
a) 16 b) 14 c) 18 d) 21
50. The Chairman of the Planning Commission in India is the
a) President b) Prime Minister
c) Minister of Planning d) Finance Minister

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Question Paper Version : C

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

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Question Paper Version : D

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

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